THE BOX

Master Documentation

An experimental computational substrate: architecture, methodology, experiments, evidence and research programme

A reference document — technical, conceptual and strategic — covering the substrate as built and the programme as conceived.

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Part I — Orientation

1. What this document is

This is an investigation into the science that could stand behind The Box.

It is not a description of the software. The architecture, the modules, the command line and the storage format are documented elsewhere, in the project's own master documentation, and they are not repeated here except where a commitment of the representation is itself scientifically significant. What is examined here is the set of ideas, principles and phenomena that the project's design either presupposes, measures, or is in a position to contribute to — and, just as importantly, the set of connections that look suggestive but do not hold up.

The framing is deliberately asymmetric. The brief for this work asked whether there is a deep connection between The Box's principles and some of the largest open questions in science, and it explicitly asked for extraordinary claims to be held to a standard of proof. The honest answer is layered, and the layering is the point:

- **Some of the substrate's commitments are already the commitments of established science.** A relational structure in which identity is derived rather than stored is a structural idea with a long history in category theory, algebraic topology and network science. A local rule system whose global statistics are tested against nulls is the central methodological move of statistical physics since the 1930s. A system in which an observable's value is required to survive a change of measurement context is exactly the operational question that Kochen–Specker made decisive in quantum foundations. None of this is exotic, and none of it is the project's invention.
- **Some of the project's own apparatus is a working instance of a research discipline**, and that discipline — not a result — is the most defensible contribution.
- **Some of the connections are genuinely surprising but not yet formal.** The substrate is a relational structure whose collective determination resembles the bulk-as-code picture in holography; it is a system whose local rules admit a limit cycle of a particular and unusual kind; it is an instrument that refuses to have a global clock, which is the posture the problem of time forces on quantum gravity. Each of these is a resemblance with content, and each is stated here as a resemblance with content — classified, and with the missing formal step named.
- **And some of the most attractive connections do not hold.** They are collected in their own chapter, because a research document that only lists successes has not been calibrated.

1.1 The register of the claims

Every substantive statement in this document carries a marker.

Marker	Meaning
`[BACKED]`	supported by established science, with a citation
`[IMPLEMENTED]`	a capability the substrate actually has, exercised by tests
`[MEASURED]`	a result recorded in the project's own experiment records
`[CLASSIFIED]`	a connection whose kind has been determined (mathematical, physical, methodological, structural, suggestive)
`[OPEN]`	an open problem in the literature that the substrate might bear on
`[PROPOSED]`	a design that does not yet exist
`[HYPOTHESIS]`	a conjecture, formulated so that it could be wrong

Marker	Meaning
`[SPECULATIVE]`	a direction with no formal content yet

Table 1.

The project's own verdict is quoted verbatim in §3.3 and it governs everything that follows: *MECHANISM EXPLAINED; NO DISCOVERY*. One phenomenon is admitted; it is explained rather than discovered. This document does not change that, and it does not soften it.

2. The substrate in one page

The Box is a computational substrate with a deliberately small set of commitments. Stated without implementation detail, because the commitments are what matters scientifically:

1. **A relational world.** Objects and relations live in one identifier space, so a relation may have a relation as one of its endpoints. Relations about relations are therefore expressible without inventing a new kind of object for the statement.
2. **Untyped labels.** Every object and relation may carry an integer label. Nothing in the system interprets a label. A reserved high range of labels is set aside for vocabulary the system did not author, so that emergent vocabulary stays distinguishable from the designer's.
3. **Derived identity.** There is no field anywhere that says "this is the same as that" or "this is the child of that". Identity classes and lineage depths are *read back* from the pattern of relations by an observer.
4. **No global clock.** Several local times coexist on relations. The counter the engine advances is an engine-side artefact and is excluded from the state's identity.
5. **Local rules, as data.** Each rule declares a predicate over a bounded neighbourhood, a selector over a bounded pool, and an effect. The engine evaluates the rules it is given; it contains no rule of its own.
6. **One site per step.** Each step, the engine enumerates the sites at which the enabled rules could fire, chooses exactly one by a stated policy, and applies that rule's effect.
7. **Measurement against nulls.** Every claim about the substrate's structure is tested against surrogate families that preserve some symmetry and destroy others, with the power of the test computed before it is run.
8. **An evidence ladder.** A verdict is resolved by one procedure with a fixed precedence, and the state in which a measurement could not be made is kept strictly distinct from the state in which the measurement refuted the candidate.

Eight commitments. The rest of this document asks, for each of them: what science already stands on it, what science has not yet been done with it, and what would have to be true for the connection to become a result.

2.1 The four ontologies, side by side

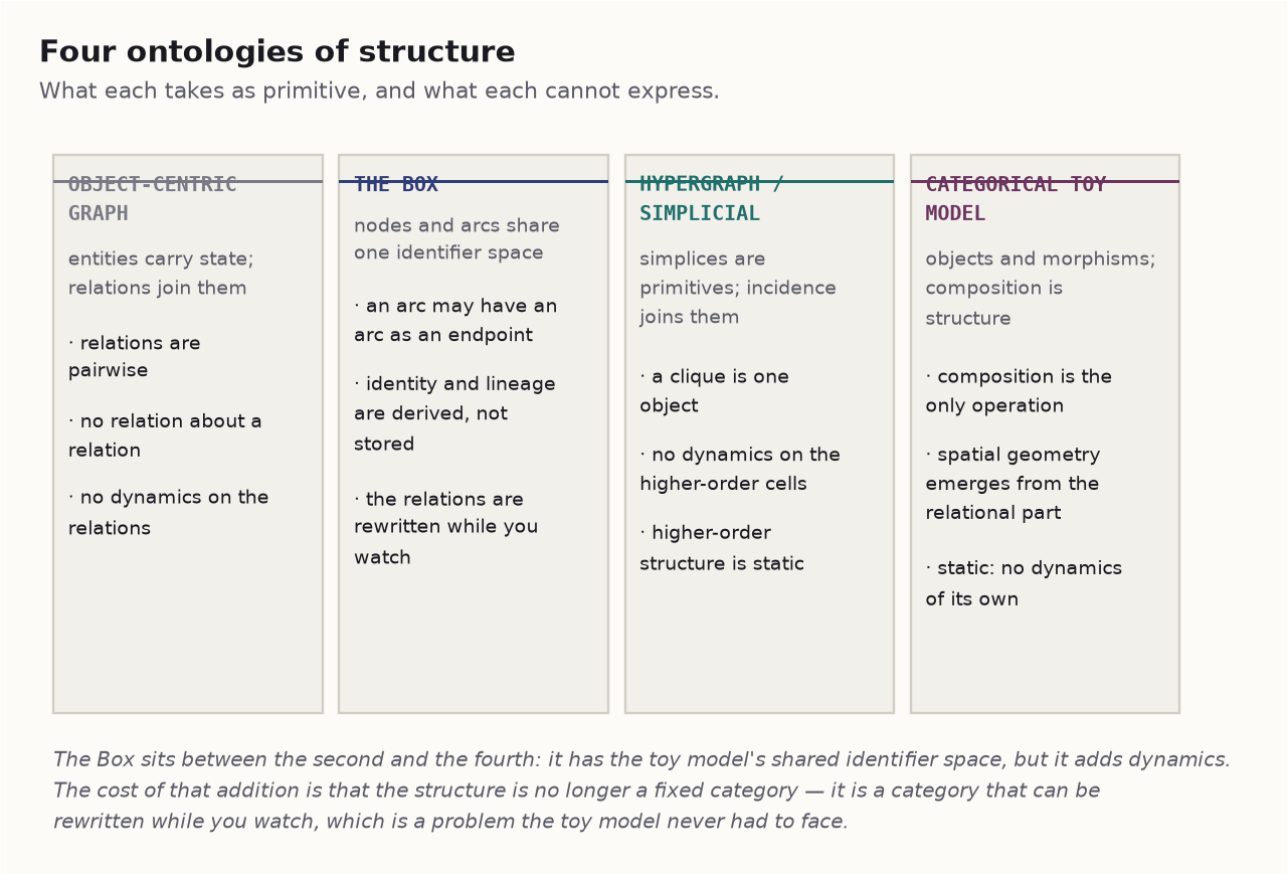


Figure 1. Four ontologies of structure

The placement of The Box in this figure is the whole of its conceptual position, and it is worth reading carefully because the differences are not cosmetic.

An **object-centric graph** takes entities as primitive and relations as derived: a relation is a labelled pair of entities. It cannot express a statement *about* a relation, and it cannot say that two relations are the same relation.

A **simplicial (higher-order) structure** takes a clique as primitive: a triangle is a single object with its own incidences. This is the standard higher-order-network move, and it is powerful — but it commits to a specific algebra of interaction that is not obviously the right one for a system whose interactions are not exhaustive.

The **categorical toy model** (Baez–Dolan) takes objects and morphisms as primitive and *composition* as the only operation. It is the most important existing formalisation of the intuition that geometry can be relational — and it is static. Its relations do not change.

The Box has the toy model's shared identifier space, but it adds dynamics: its relations are rewritten while you watch. That is a real gain and a real cost. The gain is that relations can become *subjects* of further rewriting, so a structure can change its own relational organisation. The cost is that the structure is no longer a fixed category, so the categorical apparatus — which is where the cleanest formal results about relational structure live — no longer applies directly. The gap between the second and the fourth column is the programme's real mathematical frontier, and §18.4 states it as a problem rather than a result.

3. The three commitments that make it a scientific object

3.1 Local rules, and the global laws they can produce

The question the project exists to answer is narrow and testable: **can simple local rules produce structure whose statistics are not explained by their nulls, without smuggling meaning into the rules?**

That question is not new, and the document should not pretend otherwise. It is the question of weak emergence, and the answer to it in well-understood systems is: yes, sometimes, and there are rigorous theorems. The renormisation group is the deepest of these: a global law of physics can be *derived* from a microscopic lattice model, and the derivation is exact in the sense that it identifies the fixed point the flow converges to and the exponents that characterise it. Kac's model is the cleanest small case: a global order parameter that is a theorem about the ensemble rather than a property of any particular realisation.

What is new is not the question. It is the *combination*: a substrate with relations-about-relations, identity that is derived rather than stored, several coexisting local times, and a discipline that refuses to report a result it cannot separate from its own measurement failure.

3.2 Derived identity, and the classical analogue of a "witness"

When a fusion occurs in the substrate, the relation that records it survives the removal of one of the objects it concerns. The substrate keeps a tombstone: the identifier of a removed object, still named by a surviving witness. Two substrates that differ only in their history are distinguishable, because the history is part of the structure.

This is a small design decision with a large consequence for what the system can observe. It means a trace of an event can outlive the event's participants, and that the *pattern* of relations can be read as a record of what happened. It is, in effect, an implementation of witness-based identity: the identity of two things is not asserted by a field but constituted by a relation that persists.

The same move appears, independently and much earlier, in equivalence relations in algebra, in the connected components of a graph, in the classification of states up to isomorphism, in the theory of causal sets where the order relation is the only structure, and — most pointedly for this document — in quantum error-correcting codes, where a bulk degree of freedom is defined not by itself but by its relations to a set of boundary degrees of freedom.

3.3 The discipline, which is the part that is actually finished

The project's scientific posture is stated in its own README, and it is unusual enough to be worth quoting directly:

> What it does is refuse to call anything a finding until the evidence passes gates G0–G4. Nothing here establishes new mathematics or new physics; the project's contribution is a discipline, not a result.

Translated into the vocabulary of the philosophy of science, this is a commitment to three things that are usually separated:

- **Measurement adequacy.** A statistic whose mechanism never fired is not a null result; it is an absent measurement. This is the same distinction the project draws between *not significant* and *not measurable*, and it is the same distinction that separates a failed experiment from a failed hypothesis in every mature experimental science.
- **Power before conclusion.** The null budget is computed before the run, and a configuration that cannot resolve its own significance threshold is refused rather than run and reported as negative.
- **Falsification as a first-class outcome.** One of the project's confirmed results is a hypothesis that was *falsified* 0 out of 24 confirmatory seeds, and a second that was left *inconclusive* 11 out of 24 rather than folded into the winning account.

None of this is new. It is, however, assembled into a single pipeline with an evidence ladder, a diagnosis layer that attributes rejections to named causes, and a durable store that makes every claim replayable with an explicit guarantee. Whether that assembly is worth anything to science is the question §39 and §41 attempt to answer.

4. How to read this document

Part II treats the phenomena the substrate is in a position to study. Part III asks what fundamental physics would say about a system with these commitments. Part IV asks what mathematics stands on them. Part V moves outward to complex systems, biology, computer science, artificial intelligence and cosmology. Part VI collects the connections that are less obvious than the rest and states which of them could become theorems. Part VII separates what has scientific backing from what is the project's own proposal. Part VIII sets out what could be contributed and the hypotheses that could be formulated. Part IX closes with the limits, the glossary and the bibliography.

The bibliography is at the end and every reference in it was checked against a publisher's page, an archive, or an authoritative reference work. Where an identifier could not be confirmed it is marked as unverified rather than reconstructed. Preprints are labelled as preprints.

Part II — The scientific core

5. Emergence: the word and the science

5.1 The distinction the project has to respect

The word *emergence* covers at least two different claims in the literature, and the substrate's own framing — *can simple local rules produce structure whose statistics are not explained by their nulls* — sits deliberately between them.

Weak emergence is the claim that a global property of a system follows from its local rules, but that the derivation requires the global state. The canonical instance is the renormisation group: a microscopic lattice model admits a continuum description whose long-wavelength behaviour is governed by a fixed point of the flow, and the critical exponents characterising it are independent of the microscopic details. The universality of the Ising model — the same exponents for a wide range of microscopic Hamiltonians — is the strongest established result of this kind.

Strong emergence is the claim that the global property is entailed by the local rules alone, in a way that could not have been obtained by looking at the parts. The philosophical literature on this distinction (Corning; Bedau; Gilbert & Wu; and the critical-introduction tradition that followed) is careful about a second property: in strong emergence the global property is *typically* novel in the

sense that it has no counterpart at the level of the parts, is often *irreducible* to a description from the parts, and — in the strongest claims — is *productive* in that it feeds back down.

Where the substrate stands. It has the apparatus to distinguish the two. The regime classifier reads whole trajectories rather than endpoints, precisely because a fixed point and a limit cycle have identical endpoint values; the structural search varies the *rules* rather than the substrate, precisely because a behaviour that exists only as an interaction between rules cannot be found by sampling substrates more densely. And the project has one result at the strong end — a global property, the lineage depth, that is destroyed by ablating the rule its mechanism names — and that result is *about one rule set,* not about weak emergence in general.

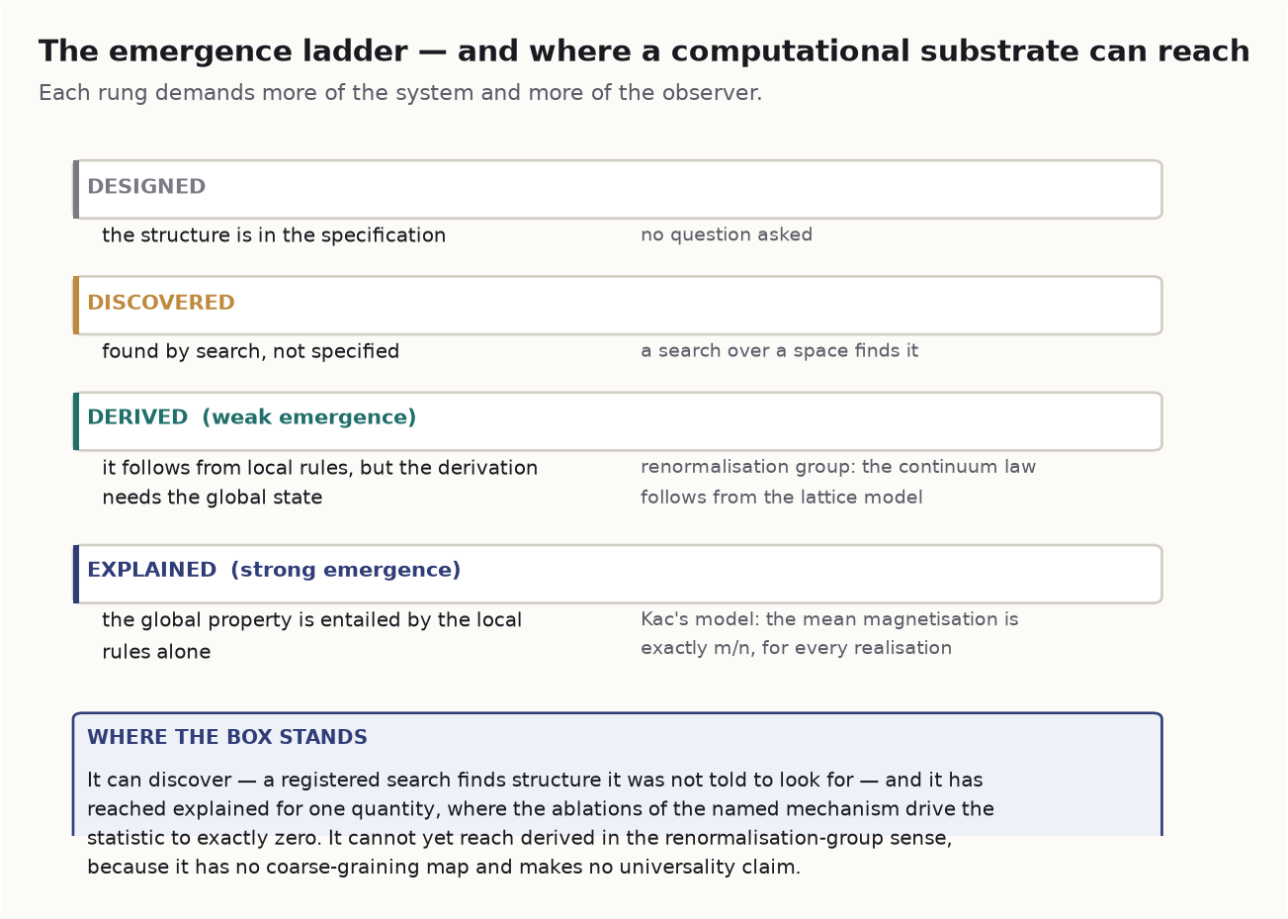


Figure 2. The emergence ladder

The ladder figure is deliberately positioned to make the uncomfortable point visible: the substrate's own ladder is *internal*. Every rung of it is a fact about the substrate and its apparatus. The external scientific ladder — §21 — adds requirements that no internal fact can satisfy.

5.2 The hardest existing case: Kac's model

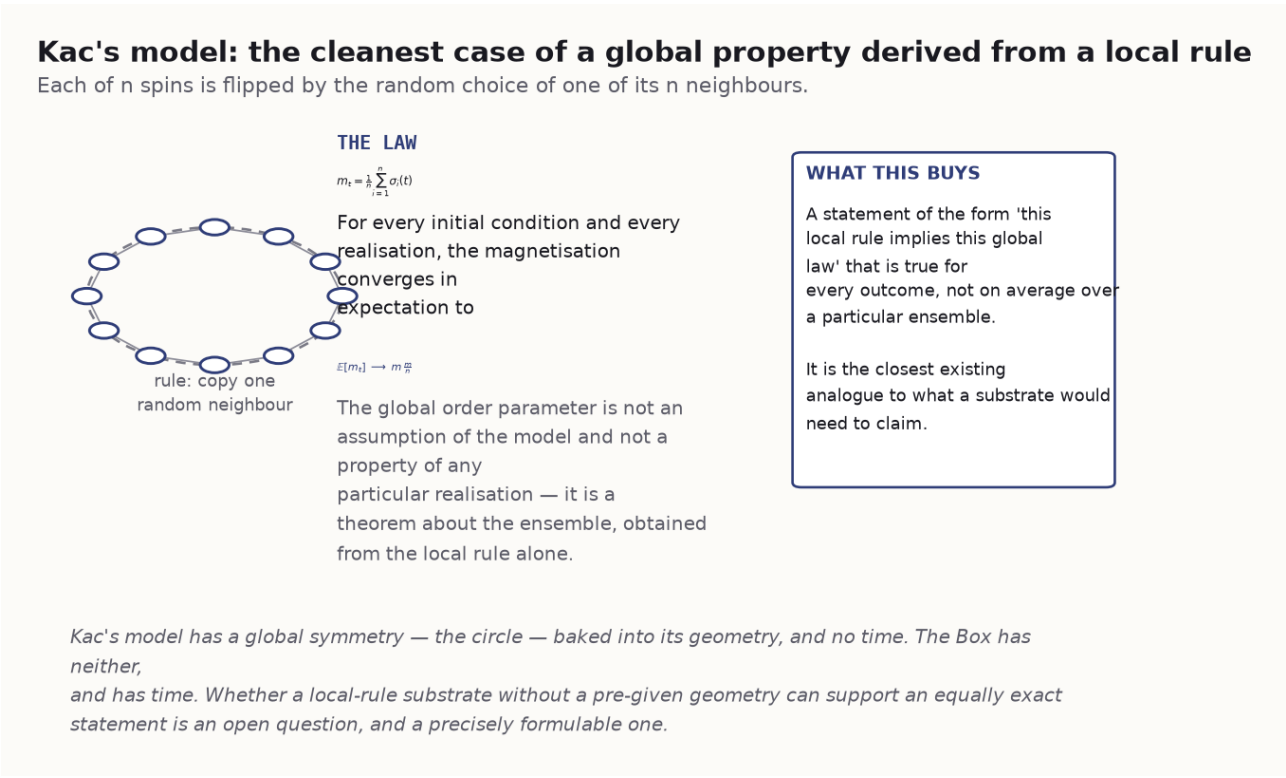


Figure 3. Kac's model

Kac's model is the cleanest small theorem of weak emergence, and it is worth stating exactly because the substrate is not Kac's model and the difference is instructive.

Take n spins on a circle. At each step every spin independently copies the value of one of its two neighbours, chosen uniformly. The magnetisation is

    m_{t} = (1/n) Σ_{i} σ_{i}(t)

Then for every initial condition and every realisation, the expectation of the magnetisation converges to m^2/n, where m is the initial excess. The binomial distribution of the counting process, its Poisson limit, and the Gaussian limit all follow in closed form.

The global order parameter is therefore not an assumption of the model and not a property of any particular realisation. It is a theorem about the ensemble, obtained from the local rule alone.

The substrate is not Kac's model, and three things are missing. Kac's rule is *linear* — a spin copies; a copy cannot create structure that was not there in a correlated way. The substrate's rules are *nonlinear* and can create objects. Kac's model has a *global* symmetry — the circle — baked into its geometry, and the substrate has no pre-given geometry. And Kac's model has *no time*: the magnetisation converges, and "converges" is a statement about the ensemble's asymptotic behaviour.

The third point is the deepest difference, and it points in an unexpected direction. A substrate *without* a pre-given geometry has to *derive* whatever global structure it exhibits, and the price is that the derivation cannot be a one-line theorem. The substrate's six rules and its four policies are a finite search space, and the question of whether the search space contains a Kac-like theorem is open — and is the sharpest formulation of the programme.

5.3 The result that changes the question: coarse-grained descriptions exist even for undecidable systems

Israeli and Goldenfeld (2006) is the single most important reference for this project, and it is the one that most changes how the question should be posed. Their result, in the terms that matter here:

- Coarse-grained descriptions can be constructed for cellular automata in **all** of Wolfram's behavioural classes — so emergence is not restricted to the edge of chaos, and the popular taxonomy is not a boundary condition on where emergence can happen.
- Some of the coarse-grainable automata are universal Turing machines and therefore undecidable — so **emergence and undecidability coexist** rather than excluding each other.
- The probability of finding a coarse-grained description approaches unity as the scale of the coarse-graining grows, because the large-scale rule is often much simpler than the primitive one. Their conclusion is that large-scale patterns are "*forced upon the system by the simplicity of the rules that govern the large-scale dynamics.*"

This is a peer-reviewed, theorem-backed version of "the system generated something that was not designed" — and, in the same paper, a theorem-backed statement of the ceiling that bounds it.

It gives the substrate something it does not currently have: a **falsifiable target**. The claim that the birth/fusion balance produces a fixed point in one case and non-settling in the other is, in this framework, a claim about the existence of an effective description at a coarse scale. The right experiment is therefore not to look for a better statistic but to search for an **effective rule set of lower descriptive complexity** that reproduces the large-scale behaviour — and to measure the complexity of the candidate effective rules against the primitive rules rather than judging "simpler" by eye.

5.4 What this changes about the project's own vocabulary

The project reserves a high range of tags for vocabulary the system did not author, so that emergent structure stays distinguishable from the designer's. Israeli and Goldenfeld give that reservation a precise reading: the question is not whether the emergent vocabulary is *new* — it cannot be, in the algorithmic-information sense of §12.4 — but whether a **description of the large-scale dynamics** exists that is much simpler than the primitive rules and was not written down. That is the only rigorous form of "the system found something I did not put there", and it is measurable.

6. Self-organisation, and the thermodynamics the substrate does not have

6.1 Prigogine's four conditions, checked honestly

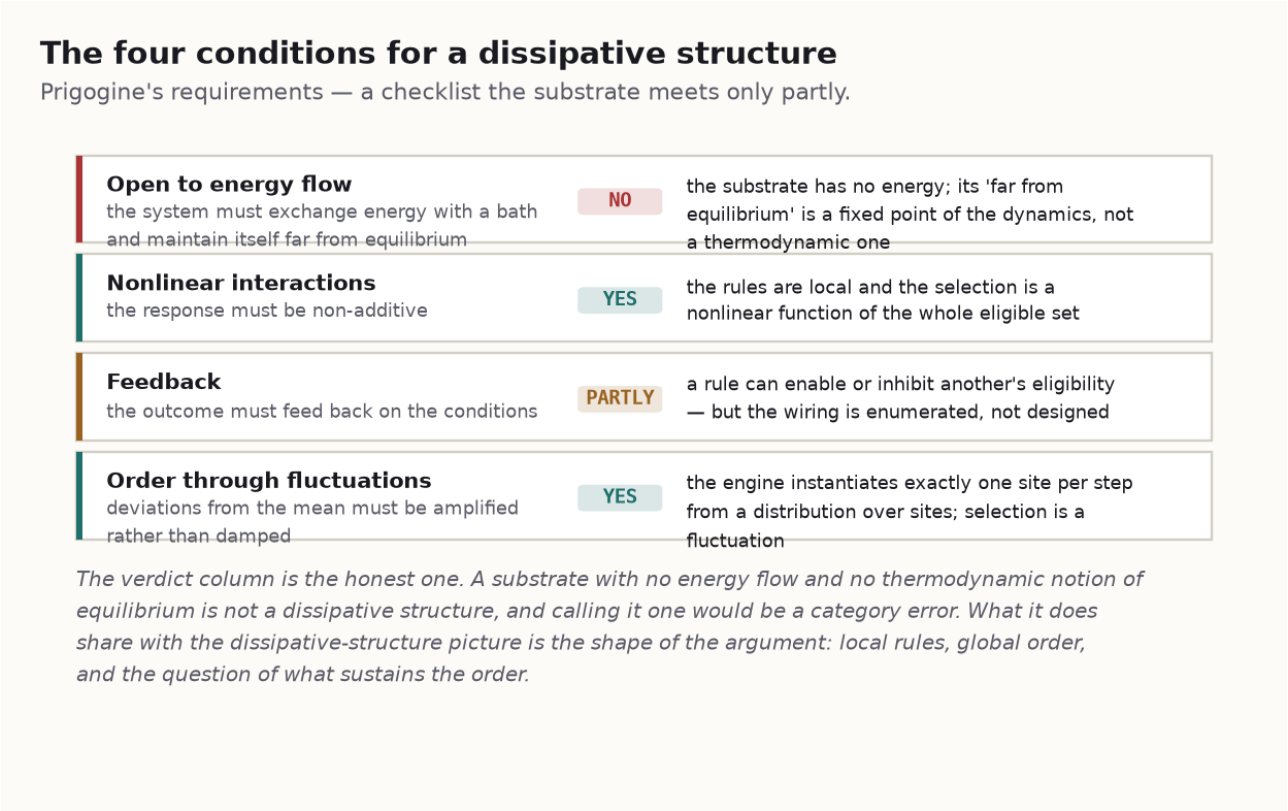


Figure 4. The four conditions for a dissipative structure

Prigogine's account of dissipative structures is the canonical statement of what is required for a system to organise itself out of equilibrium. His conditions are: an open system with an exchange of energy; nonlinear interactions; feedback; and order through fluctuations — the amplification of deviations rather than their damping. The Brusselator is the canonical minimal model: two coupled reaction variables with a driving parameter, exhibiting a Hopf bifurcation to an oscillatory limit cycle, and the oscillation is a *dissipative* structure maintained by the flow.

The figure applies the checklist to the substrate and the verdict column is deliberately unflattering. The substrate meets the second and fourth conditions and partially the third; it does not meet the first.

It does not meet the first condition, and that is not a technicality. A substrate with no energy, no entropy, and no thermodynamic notion of equilibrium is not a dissipative structure. Calling it one would be a category error. What the substrate does share with the dissipative-structure picture is the *shape* of the argument: local rules, global order, and the question of what sustains the order. And the shape is not nothing — it is the same shape that underlies Kuramoto's oscillators (a coupling threshold above which phases lock, and a critical coupling that can be computed exactly), the Watts–Strogatz small-world construction, and the self-organised criticality picture of Bak, Tang and Wiesenfeld.

6.2 Stigmergy: the substrate's own architecture, already formalised

The closest formal description of the substrate is not in physics at all. It is **stigmergy**, and it has a clean peer-reviewed lineage: Grassé's 1959 observation that the trace left by one worker's action stimulates a later action by another, through a shared medium; Theraulaz and Bonabeau's formalisation of stigmergic systems; and Heylighen's (2016) generalisation, which makes the agent unnecessary — stigmergy extends to "*agentless actions such as chemical reactions*".

The substrate's rules read relations and write relations into a shared medium. That is a stigmergic loop in the formal sense: global order arises because the trace left by each local action stimulates subsequent local actions, with no planner and no central control.

Heylighen's *strong* form of stigmergy makes a prediction the substrate can test by ablation, and this is the cheapest experiment in the whole programme: ablate entire **classes of traces** rather than individual relations, and measure whether global outcomes change. If removing a class of traces leaves the dynamics unchanged, the coordination was not stigmergic in the strong sense.

6.3 The collective behaviour literature as a template

The Kuramoto model is the cleanest worked example of emergent synchrony from local coupling, and the substrate's policy-selection problem has the same shape. In Kuramoto's model each oscillator has a phase and a natural frequency, and each is coupled to k neighbours with strength K; the self-consistency equation has a solution with a critical coupling K_c = 2γ above which phases lock and below which they do not. The global order parameter — the fraction of oscillators in the largest coherent cluster — emerges from a threshold in a local coupling law.

The substrate's selection policy has a directly comparable structure: four policies, a notion of eligibility, and a question of when the chosen sites cluster. The relevant literature for how to measure it is the higher-order network literature, where the ensemble of interest is the same one the substrate's nulls already sample.

7. Phase transitions, criticality and the regime vocabulary

7.1 The vocabulary the project already has

The substrate's regime classifier distinguishes six behaviours: transient, fixed point, limit cycle, drift, chaotic, and undetermined. This vocabulary is the substrate's own, and it is not the vocabulary of statistical mechanics — there is no order parameter, no control parameter, no exponent, no correlation length.

The figure below shows what the substrate's regime space looks like, and the three classical pictures it corresponds to.

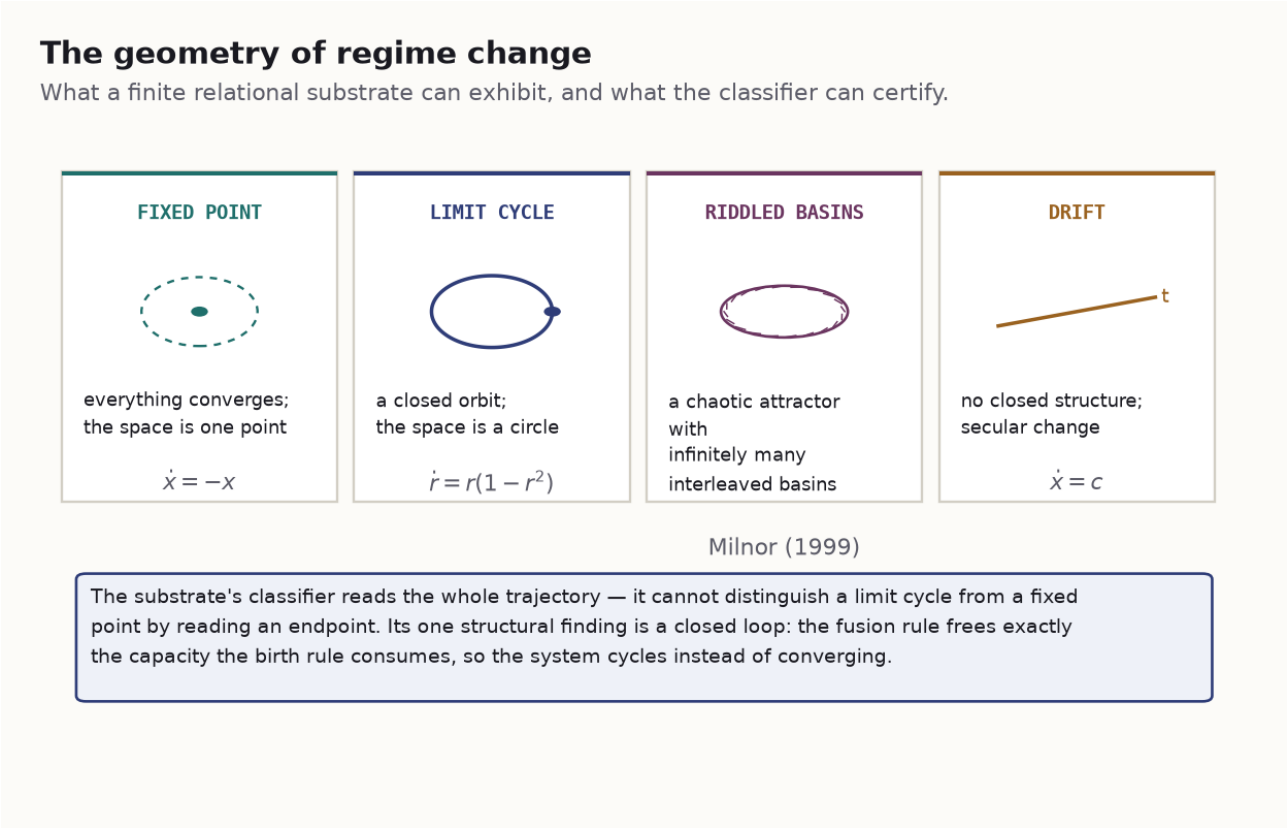


Figure 5. The geometry of regime change

7.2 What a genuine phase transition would require

For a phase transition in the substrate to be a phase transition and not merely a change of behaviour, four things would have to be supplied:

1. **An order parameter.** Something that distinguishes the two phases and takes a critical value between them. The substrate's candidates are structural (component sizes, depth distributions, the fusion fraction) and none has been shown to be one.
2. **A control parameter.** A one-parameter family through which the transition is crossed. The substrate has a genuine discrete control parameter — the *presence or absence* of the birth rule — and this is exactly the kind of parameter that produces a transition in a finite system. Whether it produces a *phase transition* is a different question.
3. **A scaling law.** A relation of the form $X \sim |\varepsilon|^{-\nu}$ with an exponent that does not depend on the microscopic details. The substrate has no ε to expand around; there is no small parameter at present.
4. **A universality class.** The decisive test, and the one the renormisation group made precise: two systems with different microscopic rules and the same exponent belong to the same class.

None of these is present. What *is* present is the observation that removing the birth rule produces a fixed point and keeping it does not, verified directly across seeds, schedulers, rule subsets and run lengths, and explained mechanistically as a capacity-coupled closed loop.

7.3 Self-organised criticality, and why the substrate is not an instance of it

Self-organised criticality — the Bak–Tang–Wiesenfeld sandpile, and its relatives — is a system with no external tuning that nonetheless drives itself to a critical point. It is the closest formal relative of the substrate's "the system finds its own operating point" intuition.

It is also contested on grounds that apply here. Paczuski's analysis and Grinstein and Roberts' work identify the absence of a rigorous mechanism: the models are not demonstrably critical, and the evidence for criticality in them is weak. The honest position is therefore that the substrate's fixed-point/non-settling distinction is *not* an instance of self-organised criticality, and that the literature's own doubts about that class of claim should be imported rather than ignored.

7.4 The one structural result, stated carefully

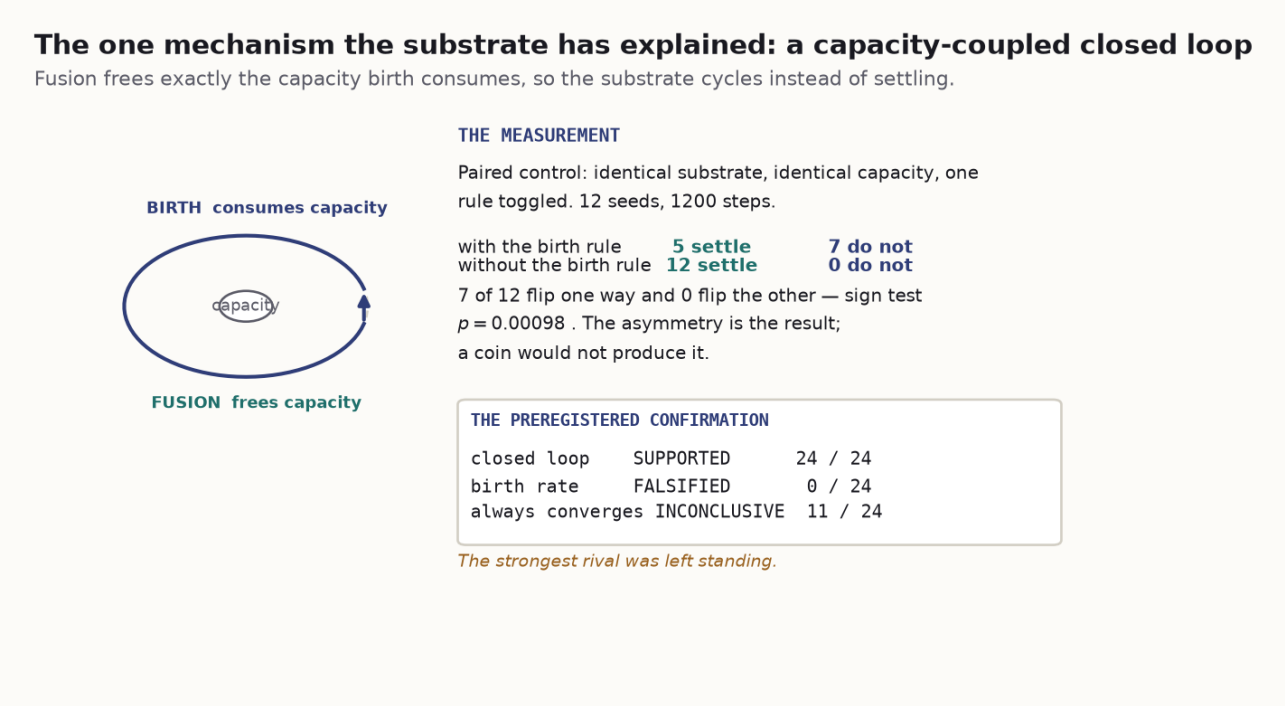


Figure 6. The capacity-coupled closed loop

The substrate's single admitted structural finding deserves to be stated with its full methodological apparatus, because the apparatus is the transferable part:

- **Observation.** With the birth rule absent the substrate reaches a fixed point, always. With it present, it never reliably settles. Verified directly across three seeds, two schedulers, all non-degenerate rule subsets and lengths from 300 to 600 steps — *independent of any p-value*.
- **Mechanism.** At the node ceiling the birth rule is blocked; the fusion rule then removes an object, freeing capacity that the birth rule immediately consumes. After saturation the firing counts were exactly equal — 17 and 17. The system is not drifting, it is cycling through a rule that refills what another rule empties. Removing the fusion rule alone settles the substrate with the birth rule fully on.
- **Paired control.** Identical substrate, identical capacity, one rule toggled; twelve seeds, 1200 steps. Seven of twelve flipped one way and none the other (sign test $p = 0.00098$). The asymmetry is the result; a coin would not produce it.
- **Preregistered confirmation** on 24 held-out seeds: the closed-loop account SUPPORTED (24/24); a birth-rate rival FALSIFIED (0/24); an always-converges rival INCONCLUSIVE (11/24)

and deliberately left standing rather than folded into the winner.

Two caveats that the literature of §12.4 forces, and that the project's own documents do not yet state.

First, the *epistemic asymmetry*: the absence of the birth rule gives a fixed point, which is a decidable, checkable fact. The presence of it gives non-settling, which — if the rule set is universal — is a potentially undecidable predicate. The two halves of the finding are not symmetric, and a reader should not treat them as one claim.

Second, and from the outside: the preregistration identifier and the protocol identifiers for this result exist only in prose within the project's Markdown. There is no stored protocol, no seed list and no re-runnable record. The project's own discipline — which requires that a mechanism claim be preregistered, ablated, and confirmed on independent seeds — would normally require exactly that artefact. **This is a gap in the project's records, not a defect in the finding, and it is the first thing that should be fixed.**

8. Invariance, symmetry and conservation

8.1 The form the project actually measures

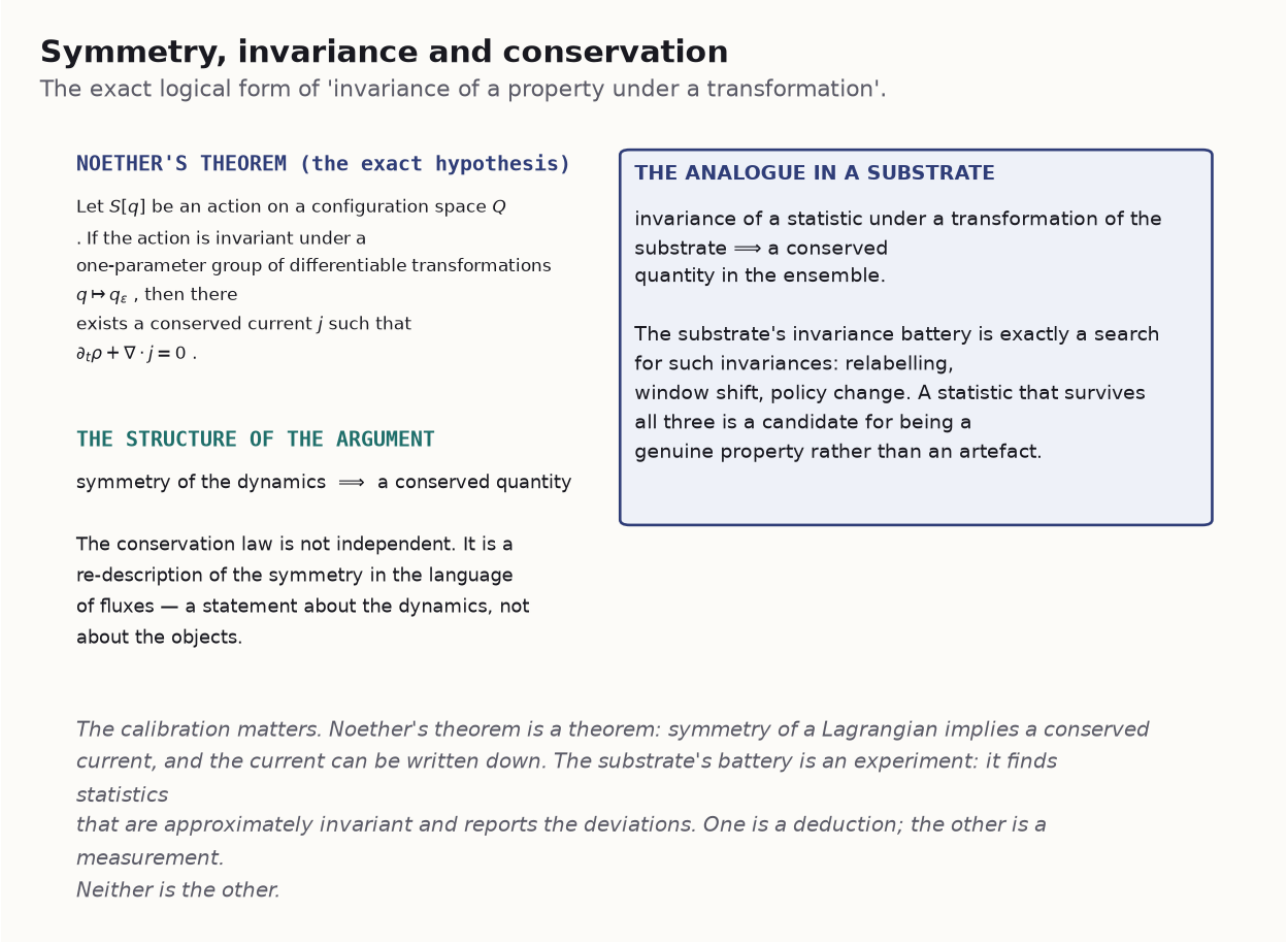


Figure 7. Symmetry, invariance and conservation

The substrate's invariance battery asks whether a statistic's value survives three changes: a relabelling of the identifiers, a shift of the observation window, and a change of the selection policy. A statistic that survives all three is a candidate for being a genuine property of the substrate rather than an artefact of the measurement.

This is, in the language of the social sciences, a **measurement-invariance** requirement, and it is the same requirement that network science imposed on its statistics once Girvan and Newman showed that a null model not matched to the data's relevant properties turns structure into artefact.

8.2 What does *not* follow, and why the distinction matters

The tempting move is to say: *invariance under a transformation implies conservation of a quantity*, and to reach for Noether's theorem. **There is no such general theorem, and Noether's theorem does not say it.**

Noether's theorem is a theorem about a differentiable symmetry of an *action functional* on a space of trajectories, yielding a momentum map whose components are conserved along solutions. The modern formulations make the hypotheses explicit — a smooth action, a smooth group action, a variational structure, and in Palais's version the Palais–Smale condition for the momentum map to exist.

The substrate has **none** of these. Its rules are discrete; there is no action functional, no trajectory space and no differentiable group. Noether does not apply to it even in weakened form.

What *does* transfer is the **method**: search for a group action under which some quantity of the dynamics is invariant, and check whether a corresponding quantity is conserved across steps. That is a search heuristic, not a theorem, and any conservation claim it produces must be verified directly in the substrate.

There is one closely related result worth knowing, and it is a preprint: Markopoulou's *Conserved quantities in background independent theories* derives conservation laws in a setting with no fixed background, which is the closest published analogue to the substrate's situation. It is a preprint and is marked as such throughout this document.

9. Information, order and the ensemble

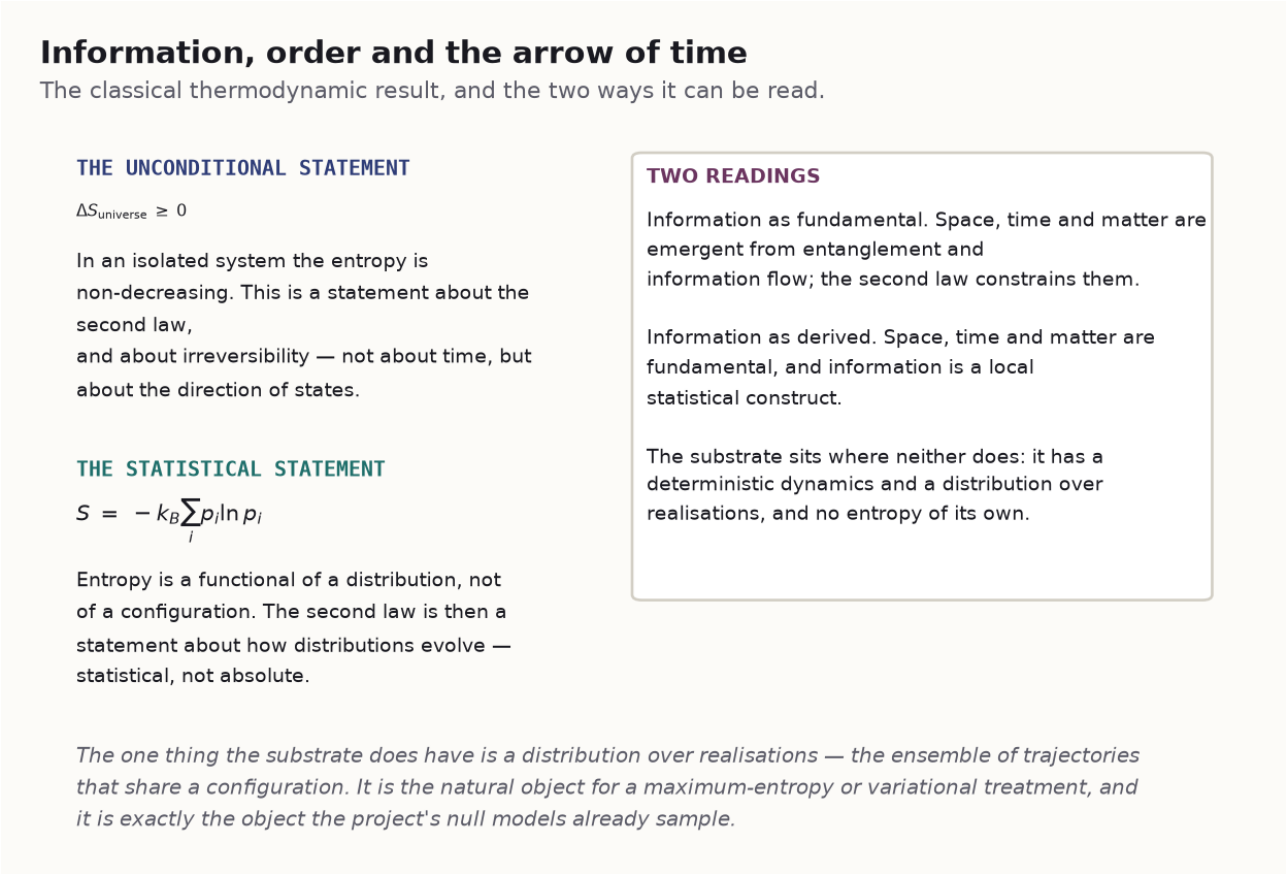


Figure 8. Information, order and the arrow of time

The unconditional statement of the second law is that the entropy of an isolated system is non-decreasing. The statistical statement, which came later and is the more useful one, is that entropy is a functional of a *distribution* rather than of a configuration, and that the second law is then a statement about how distributions evolve under a stochastic dynamics — statistical, not absolute.

The substrate's position with respect to this is precise and worth stating, because it is easy to overclaim:

- The substrate has **no entropy of its own**. It has no energy, no heat, no irreversibility.
- The substrate *does* have the object the statistical statement is about: **a distribution over realisations** — the ensemble of trajectories that share a configuration. This is exactly the object a maximum-entropy treatment would take, and it is exactly the object the project's null models already sample.
- Whether the second law has anything to say about the substrate depends on a prior question — whether the substrate's dynamics are *stochastic* at the ensemble level — and the answer is subtle. The dynamics are deterministic; the ensemble is a sampling of seeds. Whether an ensemble of deterministic trajectories obeys a monotonic entropy law is an empirical question about the substrate, not a given.

The two readings of "information is fundamental" and "information is derived" bracket the substrate's position. The substrate is a deterministic system with an ensemble of trajectories over which quantities are measured. It has the *ensemble* and it lacks the *dynamics of the ensemble*. That is a precise gap, and §42 treats it as an open hypothesis rather than a connection.

10. The ensemble as the real object

The substrate's null models sample over configurations and over realisations. This means that the *ensemble* — not the single trajectory — is the object on which its statistics are defined, and that the appropriate mathematical setting for its measurements is the one Gelman and Stern (2006) describe when they insist that "*failure to reject*" and "*failure to measure*" are different claims.

The distinction is not a private convention of the project. It is a named and load-bearing distinction in statistics, written down in a paper whose title is the argument: *Difference between* "*significant*" *and* "*not significant*" *is not merely a matter of usage*. The project imports it, and importing it is correct.

The higher-order network literature has since carried the same discipline into the setting the substrate would need — ensembles matched to the relevant properties of higher-order structure (Courtney and Bianconi; Musciotto and colleagues; Chodrow and colleagues) — and §42.7 turns this into a concrete experiment with a published protocol.

Part III — Fundamental physics

11. What the substrate is not

Before any connection is made, the exclusions should be on the record, because most of the misreadings of this project come from assuming otherwise.

The substrate is classical and discrete: integer identifiers, integer labels, real-valued payloads, integer weights, integer local times. It has no Hilbert space, no complex amplitudes, no operator algebra, no superposition, no entanglement, no measurement in the physical sense, no non-commutativity, no path integral, no energy, no entropy of its own, and no randomness in its dynamics — every "random" choice is a hash-derived function of the seed and the current state.

Therefore it is **not** quantum computation, **not** a model of quantum mechanics, **not** a theory of consciousness, free will or sentience, and **not** a claim about physical reality. The project's own directive is explicit that these are never to be claimed, and this document holds to it.

What remains is genuinely interesting, and it is worth naming precisely what it is: **a relational, discrete, deterministic dynamical system in which identity is derived rather than stored, equipped with a measurement discipline.** The question is whether such a system has anything to say to physics. The answer is: not directly, but in two specific and testable ways.

12. Relational structure, and the interpretations that take it seriously

12.1 Relational quantum mechanics

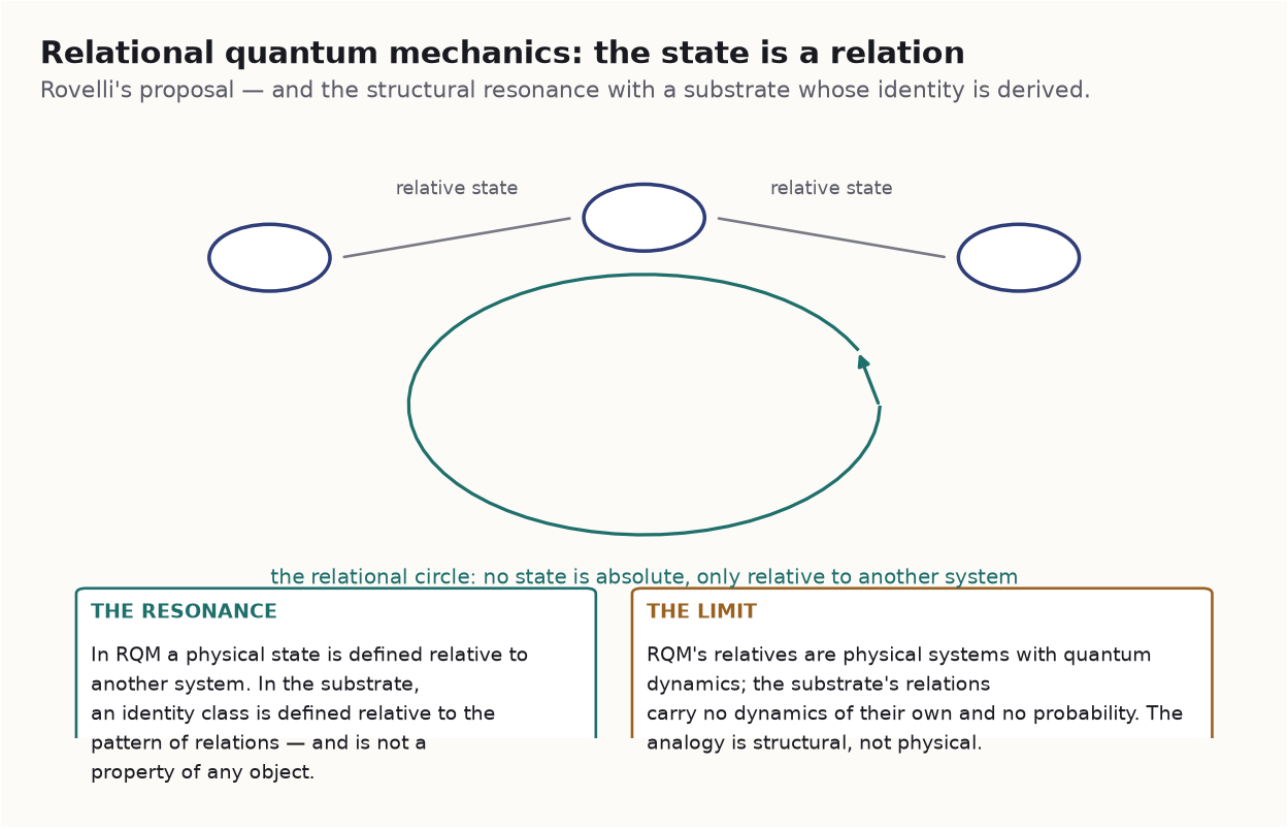


Figure 9. Relational quantum mechanics

Rovelli's relational quantum mechanics (RQM) is the most-cited position in which the physical state of a system is *relative to another system* rather than absolute. Its ontology is sparse relative facts; its motivation is the measurement problem; and its difficulties are well known and include the consistency of different observers' descriptions.

The resonance with The Box is structural and needs to be stated at the right strength. In RQM a physical state is defined relative to another system. In the substrate, an identity class is defined relative to the pattern of relations — and is not a property of any member of it. In both cases, the relational structure is what carries the determination.

But the difference is decisive and must not be glossed. RQM's relatives are physical systems with quantum dynamics, and the relations between them are probabilistic. The substrate's relations carry no dynamics of their own and no probability. The analogy is structural; it is not physical, and no result transfers.

What *does* transfer is a much better formalisation than RQM's, which the substrate resembles far more closely than the popular literature suggests.

12.2 The categorical formalisation: causal structure without a background

The literature that actually treats relations as primitives is the categorical one, and the substrate's architecture is recognisably in that family.

- **Poulin (2006)**, *A relational formulation of quantum theory*, gives a purely relational description of quantum theory in which reference systems are promoted to dynamical entities — the "toy model" genre the substrate most resembles.
- **Kissinger and Uijlen (2019)**, *A categorical semantics for causal structure*, give causal structure a semantics in a dagger-compact category, with **no background time and no metric**.

This is the closest published formalisation to what the substrate does when it derives identity and lineage depth from relation patterns instead of storing them.

- **Coecke and Lal** (preprint) define causal structure and time-asymmetry in a categorical language over relational data alone.
- **Baez (1996)**, *Spin networks in gauge theory*, recasts geometry as a combinatorial structure on edges — the relational substrate of loop quantum gravity.

And the most important formal point for this document is the one the research dossier makes: the papers that actually *derive* time are less cited than the ones that merely state that time is absent. Kauffman and Smolin, and Markopoulou and Smolin (2004), define the causal order on a relational structure of states by the algebra of local transition operators and derive locality intrinsically. The substrate makes the same formal move — deriving a global property from a relation space rather than assuming it — and does it for both time-like ordering *and* identity, from one relation space.

This is the strongest mathematical resonance in the document, and it is classified `[CLASSIFIED: MATHEMATICAL]`. The shared formal object is: *a relation space on which a global order or a global partition is defined by an algebra over local structure, with no background geometry assumed.*

12.3 Where the substrate differs importantly

The categorical programme is about *static* relational structure. The substrate's relations are rewritten. That is a real difference in kind, not degree: in a dagger-compact category the composition is fixed, and every result about the category is a statement about a fixed structure. The substrate's composition is *dynamic*, and the question of whether the categorical apparatus survives a rewriting of the morphisms is precisely the open question.

There is a second difference, which is more subtle and possibly more interesting: the substrate's relations can have relations as endpoints. In the categorical formulations the objects are the things and the morphisms are the relations, and that distinction is maintained. The substrate collapses it. Whether the collapse is an advantage or a problem is not known; what is known is that it is the place where the substrate is unlike everything in the literature.

13. Contextuality: the measurement problem, restated in the substrate's own language

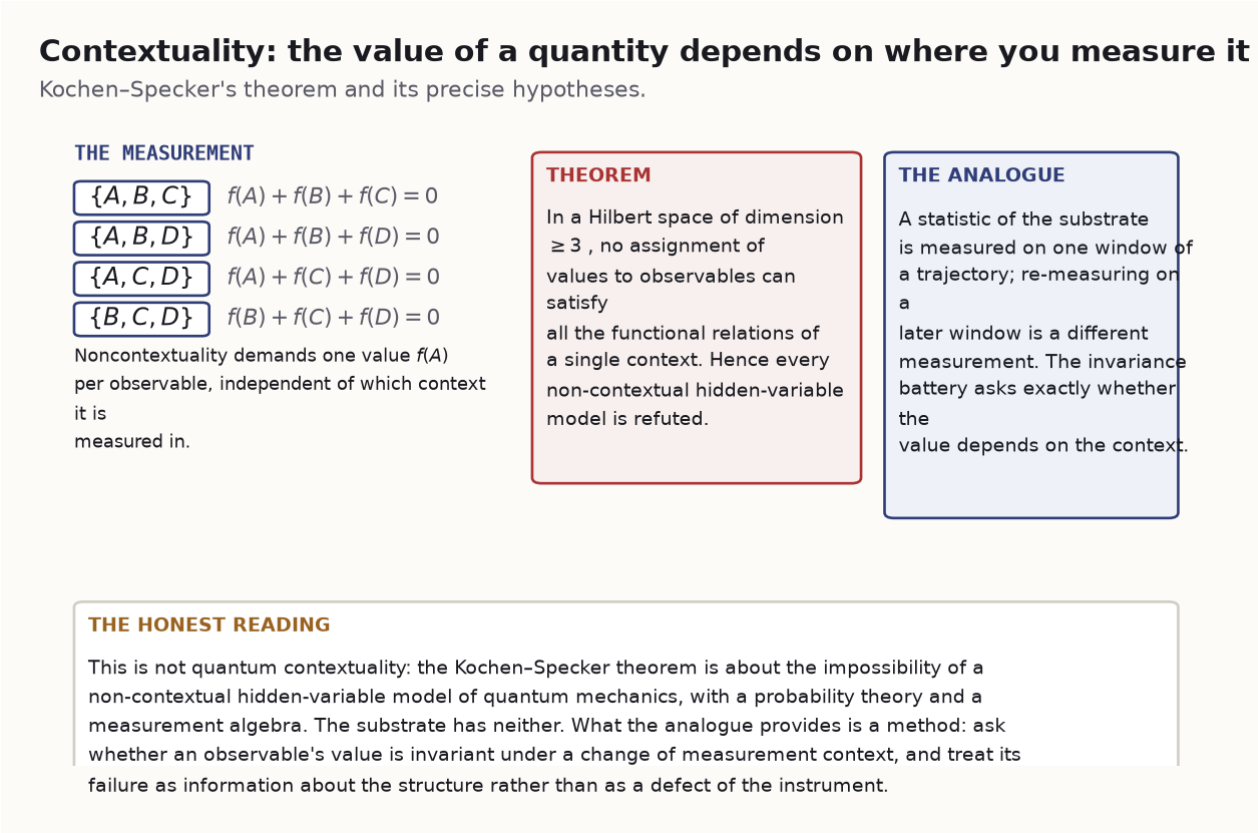


Figure 10. Contextuality

13.1 The theorem

The Kochen–Specker theorem is the statement that in a Hilbert space of dimension at least three, no assignment of values to observables can satisfy all the functional relations of a single measurement context. Its consequence for physics is that **every hidden-variable theory reproducing quantum mechanics must be contextual** — the value of an observable depends on which set of compatible observables is measured alongside it.

Spekkens' operational formulation sharpened the concept: noncontextuality is the conjunction of *preparation* noncontextuality and *measurement* noncontextuality, and the two are operationally equivalent in quantum theory — a preparation that is operationally equivalent to a mixture must give the same statistics. Cabello, Severini and Winter constructed contextuality scenarios as *hypergraphs*, with the hypergraph encoding a set of measurement procedures rather than a physical network of interacting systems; Kunjwal's analysis addresses the sharpness of contextuality.

One detail matters for this document. The hypergraph in that literature is a combinatorial encoding of an *experiment*. It is not a network of interacting agents. That distinction will matter again in §19, where the substrate's own hypergraph-shaped structure is compared with the network-science one, and in §17.1.

13.2 The substrate's version of the question

The substrate measures a statistic on one window of a trajectory and re-measures it on a later window. These are two different measurements of the same trajectory, and the invariance battery asks precisely whether the value depends on the context.

That is a genuine structural parallel, and it is the most direct one in this document: **the substrate's window-invariance probe is a contextuality test in Spekkens's sense, applied to a classical discrete system.**

It is also a much weaker claim than the quantum theorem, and the difference should be stated precisely. The Kochen–Specker theorem is about the impossibility of a non-contextual *hidden-variable model of quantum mechanics*, in a system with a probability theory, a measurement algebra, and specific algebraic constraints among contexts. The substrate has none of these. What it has is a *method* — ask whether an observable's value is invariant under a change of measurement context — and the discipline to treat its failure as information about the structure rather than as a defect of the instrument.

Classification: `[CLASSIFIED: METHODOLOGICAL]`**.** The method transfers; the theorem does not, and no claim of quantum contextuality is made or implied.

13.3 Why this is nevertheless worth pursuing

The inverse contextuality programme — using contextuality as a *resource* rather than an obstruction — is a live area, and the substrate's framing sits closer to it than the obstruction framing: the substrate is not asking whether a hidden-variable model exists, it is asking whether a statistic's value is stable across the contexts in which it could be measured, and it treats instability as a finding.

14. Emergent spacetime

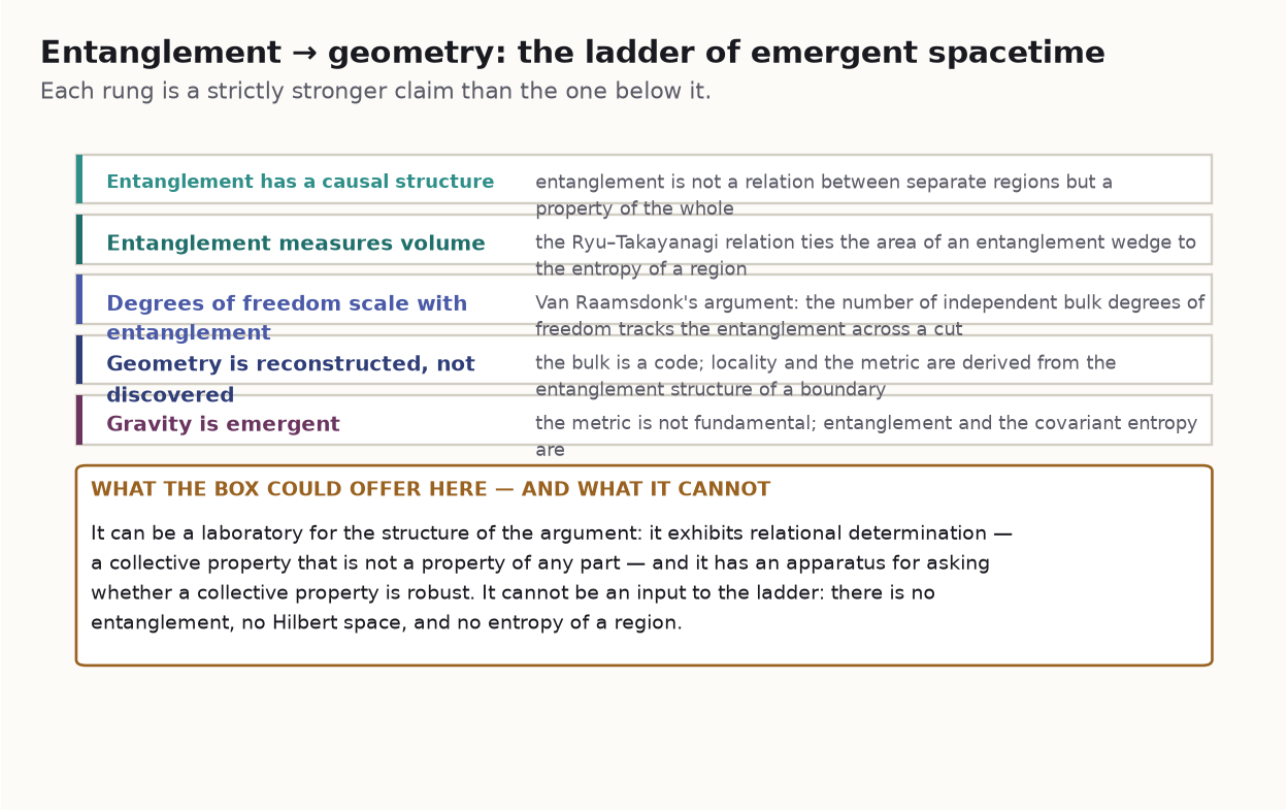


Figure 11. The entanglement-to-geometry ladder

The programme in which space and time are *derived* rather than assumed has a well-defined ladder, and each rung is strictly stronger than the one below.

14.1 The ladder

Entanglement has a causal structure. Entanglement is not a relation between separate regions but a property of the whole.

Entanglement measures volume. The Ryu–Takayanagi relation ties the area of an entanglement wedge to the entropy of a boundary region.

Degrees of freedom scale with entanglement. Van Raamsdonk's argument (2010) is that the number of independent bulk degrees of freedom tracks the entanglement across a cut, so that a connected bulk geometry is not the generic outcome of the construction but something that has to be paid for in entanglement. Swingle (2018) reviews the programme.

Geometry is reconstructed, not discovered. The bulk is a code; locality and the metric are derived from the entanglement structure of a boundary.

Gravity is emergent. The metric is not fundamental; entanglement and the covariant entropy are. Godet and Marteau (2019) give the flat-spacetime version with its assumptions stated.

Jacobson's (2016) variational argument is the most economical statement of the underlying idea: a variational principle on entanglement entropy reproduces Einstein's equations, so geometry enters through entanglement first.

14.2 What the substrate can and cannot contribute here

Cannot. It has no entanglement, no Hilbert space, no entropy of a region, no horizon, no large-N limit. Every premise of the ladder is absent, and the inference does not survive the transfer. Applying entropic gravity to the substrate's "growth" is a **category error**, and the research dossier is explicit that the *shape* of the argument is present on both sides while the inference is not.

Can. The substrate is a laboratory for the *structure* of the ladder's first rung — the claim that a global property can be a property of the whole and of nothing in it. It exhibits relational determination; it has a vocabulary for redundancy; and it has an apparatus for asking whether a collective property is robust. That is a contribution to the methodology of the programme, not to its physics.

There is one place where the substrate is more than a methodological analogue, and it is not in the ladder but beside it.

14.3 Quantum Graphity: the closest relative, and what it shows

Quantum Graphity (Konopka, Markopoulou and Severini, 2008) starts from a discrete state on a graph, evolves it with local rules only, and derives *both* the continuous spacetime *and its locality* — locality is a derived quantity, not an input. Hamma, Markopoulou, Lloyd, Caravelli, Severini and Markström (2010) then give an exactly solvable local model whose evolving graph supplies the emergent geometry, which is a rigorous model rather than an analogy.

This is the closest existing relative of The Box, and it is worth being precise about how close:

	Quantum Graphity	The Box
State	discrete, on a graph	discrete, relational, order ≥ 2
Rules	local	local, declarative, as data
Emergent	spacetime **and locality**	identity, lineage, regime
Geometry	derived and central	not present

	Quantum Graphity	The Box
Verified	exactly solvable variant	one mechanism, preregistered confirmation

Table 2.

The gap in the third row is the important one, and it is the direction in which the substrate is *behind* Graphity: the substrate derives identity and lineage but has no geometry at all. §16 asks what would be needed for it to derive one.

15. Holography: the strongest structural analogy available

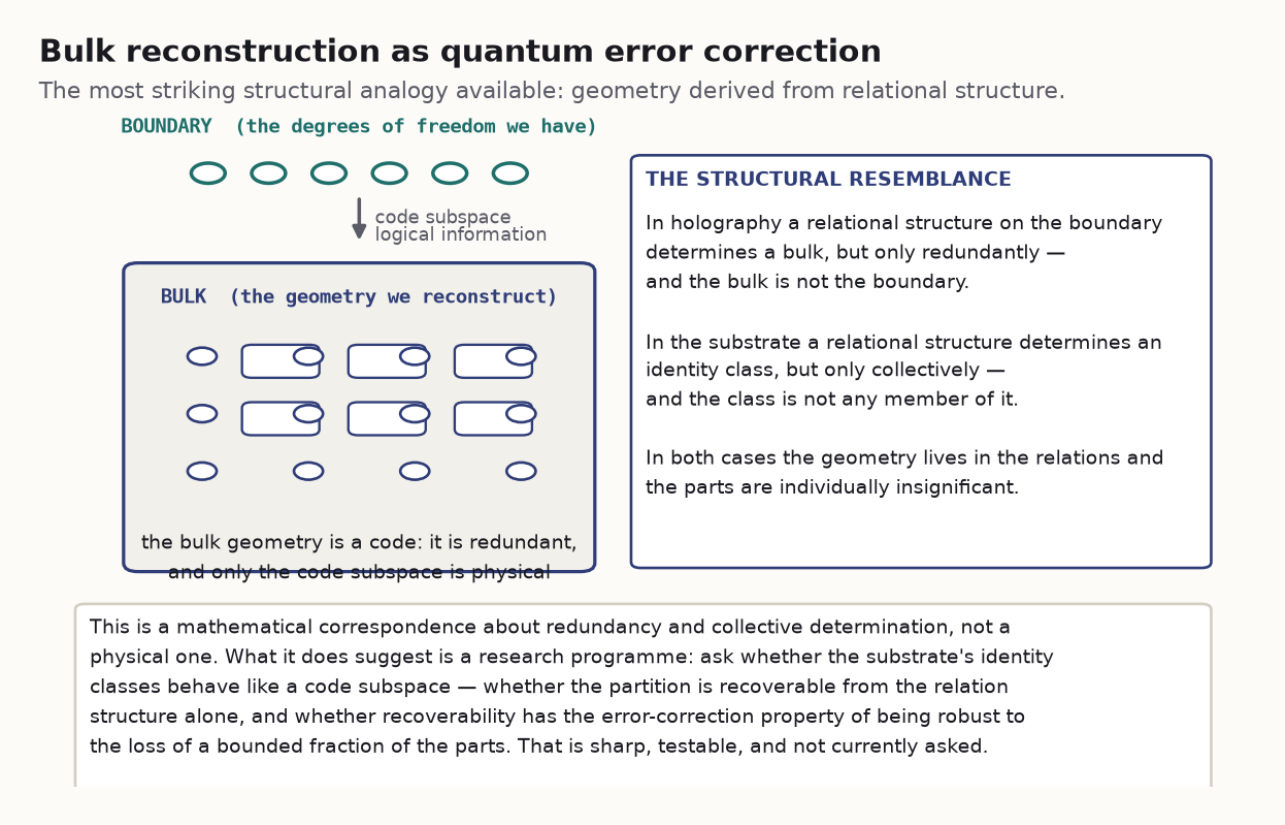


Figure 12. Bulk reconstruction as quantum error correction

15.1 The statement

In the AdS/CFT programme, bulk locality is identified with quantum error correction: the bulk is reconstructed from the boundary, and locality is *derived* from a nonlocal encoding. Almheiri, Dong and Harlow (2015) make the identification; Penington (2020) and Cotler, Hayden, Penington, Salton, Swingle and Walter (2019) give the reconstruction as an operator-algebraic recovery map; Cotler, Penington and Ranard (2020) show that bulk locality follows from the structure of the boundary Hamiltonian's spectrum; and Hayden and Penington (2017) state the exact hypothesis under which a bulk operator can be reconstructed from the boundary at all. Almheiri, Hartman, Maldacena, Shaghoulian and Tajdini (2021) review the island formula, in which the bulk region is determined by boundary entanglement. Lévay and Holweck (2019) give a *finite* geometric toy model of spacetime as an error-correcting code — a small, checkable instance of the idea.

15.2 The resemblance

The resemblance is genuinely close, and it is a resemblance about *redundancy and collective determination*:

- In holography a relational structure on the boundary determines a bulk — but only **redundantly**, and the bulk is **not** the boundary.
- In the substrate a relational structure determines an identity class — but only **collectively**, and the class is **not** any member of it.

In both cases the geometry lives in the relations and the parts are individually insignificant. This is the most striking formal parallel available, and it is classified `[CLASSIFIED: STRUCTURAL]`.

15.3 Why it is not yet a mathematical connection, and what would make it one

The parallel is currently an *architecture* resemblance. Two things would have to be supplied to turn it into a theorem-like statement.

First, a definition of what "reconstruction" means for the substrate. The holographic notion is precise: a bulk operator is reconstructed from a boundary region when the code subspace admits a recovery map. The substrate's identity classes are equivalence classes of a relation pattern; whether the partition is *recoverable* from a subset of the relations — and *robustly* recoverable in the sense that losing a bounded fraction of the parts leaves the partition intact — is a well-posed question with a natural metric, and it is **not currently asked anywhere**.

Second, a notion of locality to recover. The holographic programme derives locality because it has a metric; the substrate has none. §16 states what would be needed.

A boundary case worth stating plainly. The substrate's tombstone set — the identifiers of removed objects, retained because a surviving witness still names them — is *not* an error-correcting code, and the resemblance should not be pushed. It is a bookkeeping device that makes two substrates with different histories distinguishable. It has no recovery property, no redundancy budget, and no error model. The resemblance that matters is at the level of the *identity partition*, not at the level of the tombstone.

16. The problem of time, and the two ways to derive an order

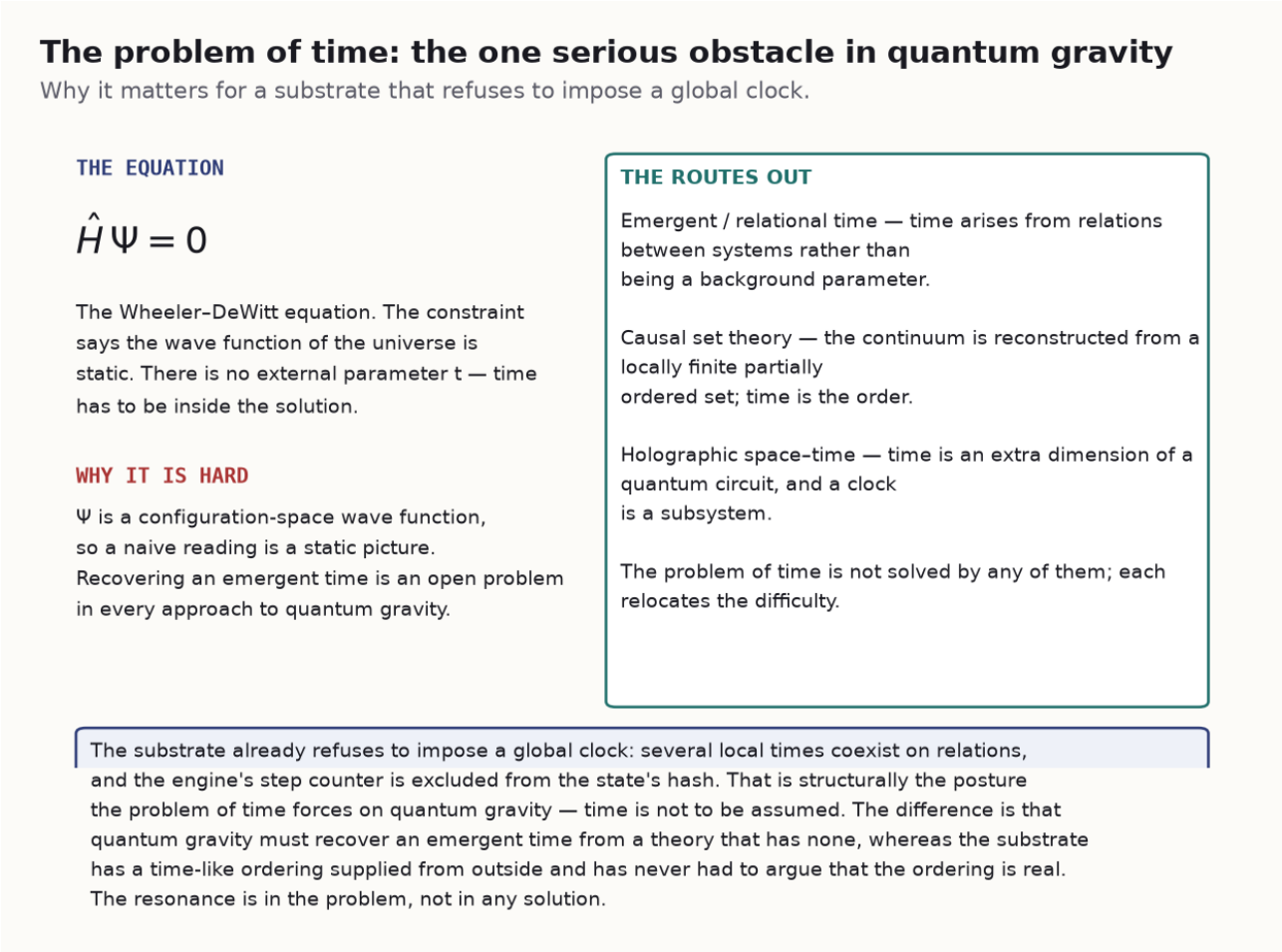


Figure 13. The problem of time

16.1 The obstacle

The Wheeler–DeWitt equation, $\hat{\mathbf{H}}\Psi = 0$, is the Hamiltonian constraint of general relativity. The wave function of the universe is static; there is no external parameter t; and time has to be somewhere inside the solution. Every approach to quantum gravity faces this, and the standard literature is largely a statement that time is *absent* (Isham; Kuchař; Barbour; Ashtekar and Isham).

16.2 The resonance, and its exact limit

The substrate already refuses to impose a global clock. Several local times coexist on relations; the engine's step counter is an engine-side artefact excluded from the state's identity. That is *structurally* the posture the problem of time forces on quantum gravity — time is not to be assumed.

The difference is that quantum gravity must **recover** an emergent time from a theory that has none, whereas the substrate has a time-like ordering supplied from outside and has never had to argue that the ordering is real. The resonance is in the *problem*, not in any solution.

16.3 The second derivation: causal structure from relations

The papers that derive time rather than stating its absence are the ones that matter, and they are the ones the substrate most resembles.

Markopoulou and Smolin (2004) reconstruct quantum mechanics from a theory of relational, non-geometric states, with physical laws appearing as constraints on the causal evolution of a graph; the causal order is defined on a relational structure of states by an algebra of local transition operators, and locality is derived intrinsically. Kauffman and Smolin pursue the same programme.

Quantum Graphity and its exactly solvable variant (§14.3) derive spacetime and locality from a graph state under local rules.

Causal set theory derives a Lorentzian metric from a locally finite partially ordered set with a measure (Bombelli, Lee, Meyer and Sorkin, 1987), proves that the continuum is recovered (Rideout and Sorkin, 2001), and shows how the emergent metric is fixed up to a global conformal factor and how the dynamics respect the emergent causal structure (Dowker, Imambaccus, Owens, Sorkin and Zalel, 2020). Sorkin and Yazdi (2017) compute an entropy on a causal set — the information-to-geometry programme run on relational-only data. Diamantini and Trugenberger (2017) exhibit a network-level order parameter detecting the onset of geometric order in a discrete model, which makes the emergence of geometry *measurable*.

Barbour (1974) gives the relational-only premise in a peer-reviewed letter: a framework in which the dynamical law is expressed only in terms of the relative distances between entities.

16.4 The hypothesis this generates, and the falsifiable test

The substrate's relations are *not* an order: several local times coexist with no stated consistency. The causal-set machinery requires **transitivity**, **local finiteness** and a **measure**, and the substrate's "several local times coexist" is exactly the condition under which that machinery would **fail**.

So the connection is strong in form and fragile in hypothesis — which is precisely why it belongs both on the list of promising connections and on the list of things that could be falsified. And the test is cheap and decisive:

> **Build the time-augmented relation graph of the substrate and test it for cycles.**

If it is a partial order, the substrate has *derived time* in the exact sense of the Markopoulou–Smolin and causal-set papers, and the correspondence becomes a theorem-shaped claim. If it is not, the "no global clock" commitment is an obstruction rather than an affordance, and the causal-structure programme is closed to it.

This is the highest-value single experiment the literature suggests, and it is cheap.

16.5 A caution that the literature forces

The substrate's "no global clock" is **not** the same as relativity's "no preferred frame", and **not** the same as causal sets' "no global time function". Relativity has a global time function in each inertial frame and a light cone. Causal sets are locally finite posets — a genuine partial order with transitivity — and their no-global-time-function property follows from the order being non-totally ordered, not from an absence of structure. The substrate may have neither. The connection is

`[CLASSIFIED: STRUCTURAL]` at best, and contingent on a transitivity test that has not been run.

17. A metric from relations, without holography

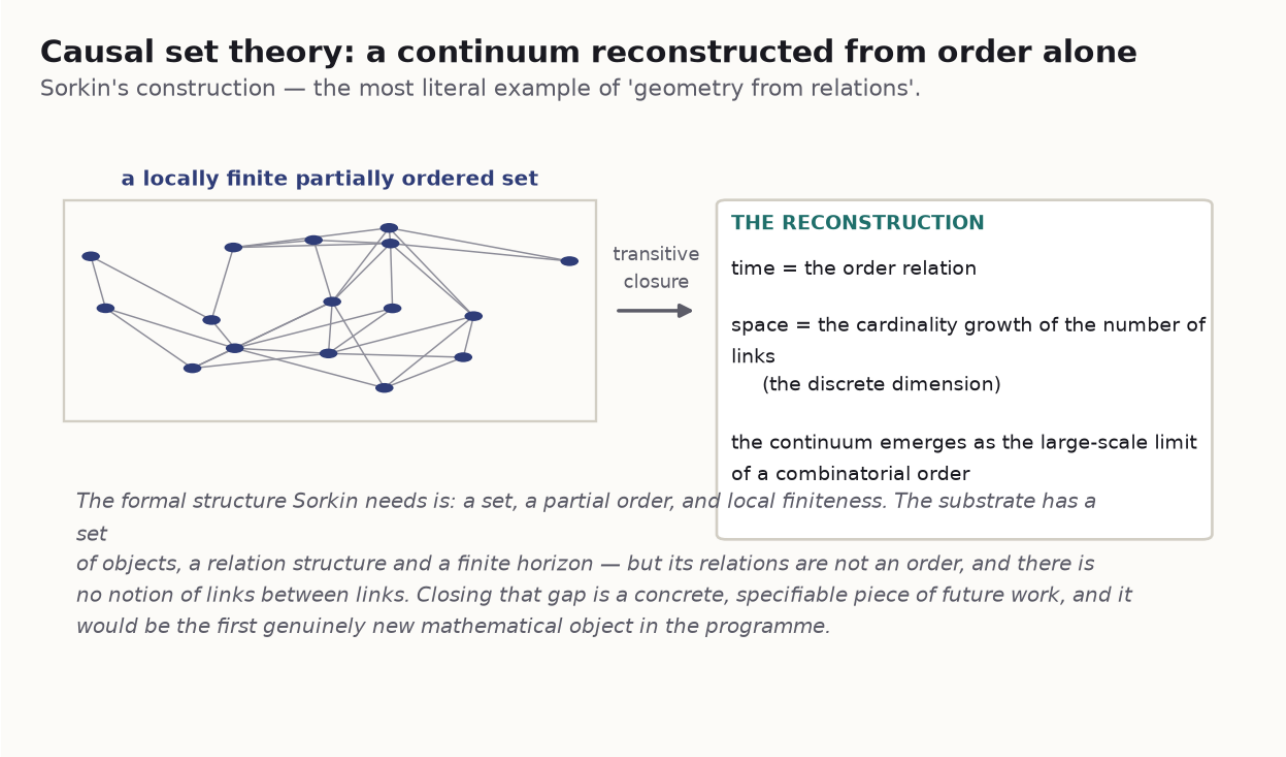


Figure 14. Causal set theory

The most attractive feature of the causal-set programme for a computational substrate is that its central claim is a **theorem** rather than a conjecture: a locally finite partially ordered set with a measure supports a continuum Lorentzian geometry, and the continuum limit can be proved to exist.

And there is a second route that is not quantum gravity at all and is, for a computational substrate, more immediately usable: **network geometry**. Bianconi and Rahmede (2017) derive a hyperbolic metric from relational data plus a generative ensemble. Millán-Torres, Torres and Bianconi (2019) read a spectral dimension off a relational structure.

Both require only three things the substrate already has: relational data, an ensemble, and a functional of the structure. Neither requires a Hilbert space, a horizon, or a large-N limit.

The physics of holography gives the ambition; the mathematics of network geometry gives the toolbox. And the substrate has a third route it has not yet tried, which is the one the causal-set papers suggest: define a **spacing** from the size of relational neighbourhoods, test whether it satisfies a triangle-inequality-like constraint, and check whether the recovered dimension is stable across scales. If it is, the substrate has a metric derived from relations in the exact sense of Sorkin — and the holographic literature becomes an ambition rather than a necessity.

Part IV — Mathematics

18. Categories, toposes, and the toy model

18.1 The strongest formal support for "geometry can be relational"

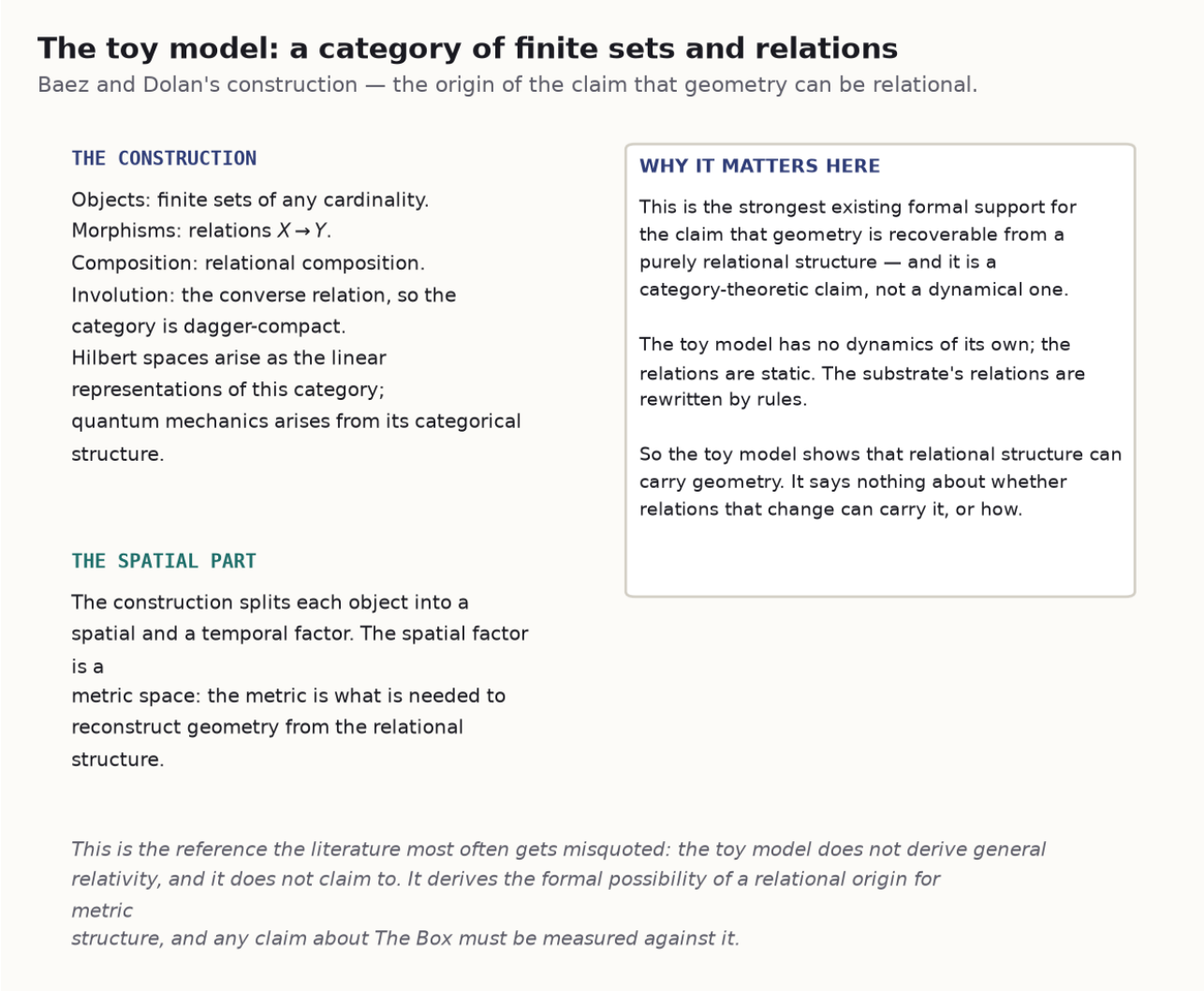


Figure 15. The toy model

The **toy model** — Baez and Dolan's categorical quantum mechanics — is the reference work for the intuition that the substrate shares. Its construction:

- **Objects:** finite sets of any cardinality.
- **Morphisms:** relations $X \to Y$.
- **Composition:** relational composition.
- **Involution:** the converse relation, so the category is dagger-compact.
- **Hilbert spaces** arise as the linear representations of this category, and quantum mechanics from its categorical structure.

The construction splits each object into a *spatial* and a *temporal* factor. The spatial factor is a metric space, and the metric is exactly the information needed to reconstruct geometry from the relational structure. The temporal factor is where the dynamics go.

Two points about this that bear directly on The Box.

First, the toy model is the formal source of the claim that **a relational origin for geometry is not a metaphor**. It is a construction, in a category with a dagger, in which spatial structure is reconstructed from relations. That is a much stronger statement than "geometry and relation feel connected", and it is the yardstick against which any claim about The Box and geometry must be measured.

Second, and less comfortably: the toy model is **static**. Its relations do not change. The substrate's relations are rewritten while you watch. So the substrate is in the position of wanting to do what the toy model does, but with the extra difficulty that the structure it would have to be categorised over is itself in motion.

18.2 The topos, and what it would mean

A topos is a category with finite limits, arbitrary joins, an exponential, and a logic — the internal logic of a topos is intuitionistic, and the topos is a categorified model of set theory. The toy model *is* a topos, and its internal logic is the logic of sets.

The consequence that matters here: **in a topos, a proposition is not a truth-value but a subobject of the terminal object.** Whether something is true is determined by whether it is *inhabited* — and inhabitation is a question about the existence of a global element, which is a question about the structure.

This is the deepest available formal analogue of what the substrate does with identity. The substrate asks: is there a configuration that witnesses this identity class? The topos asks: is this subobject of the terminal object inhabited? Both replace a truth-value question with a question about the existence of a witness in the structure.

18.3 The gap, stated as a research problem

The categorical apparatus is about fixed structure; the substrate's is not. And there is a second, sharper gap that the dossier makes explicit: **there is no verified primary source stating the categorical toy model's relationship to causal structure in the form the substrate would need.** The closest verified statement — Kissinger and Uijlen's categorical semantics for causal structure — is a different construction, in a dagger-compact category, and it defines causality *on* relational data rather than deriving it *from* rewriting.

So the honest position is:

> The substrate's architecture is in the family of the toy model, and the toy model is the strongest formal support for the claim that geometry can be relational. But the substrate does not do what the toy model does, the toy model does not do what the substrate does, and the literature does not currently connect them. `[PROPOSED]`

19. Higher-order structure: simplicial versus relational

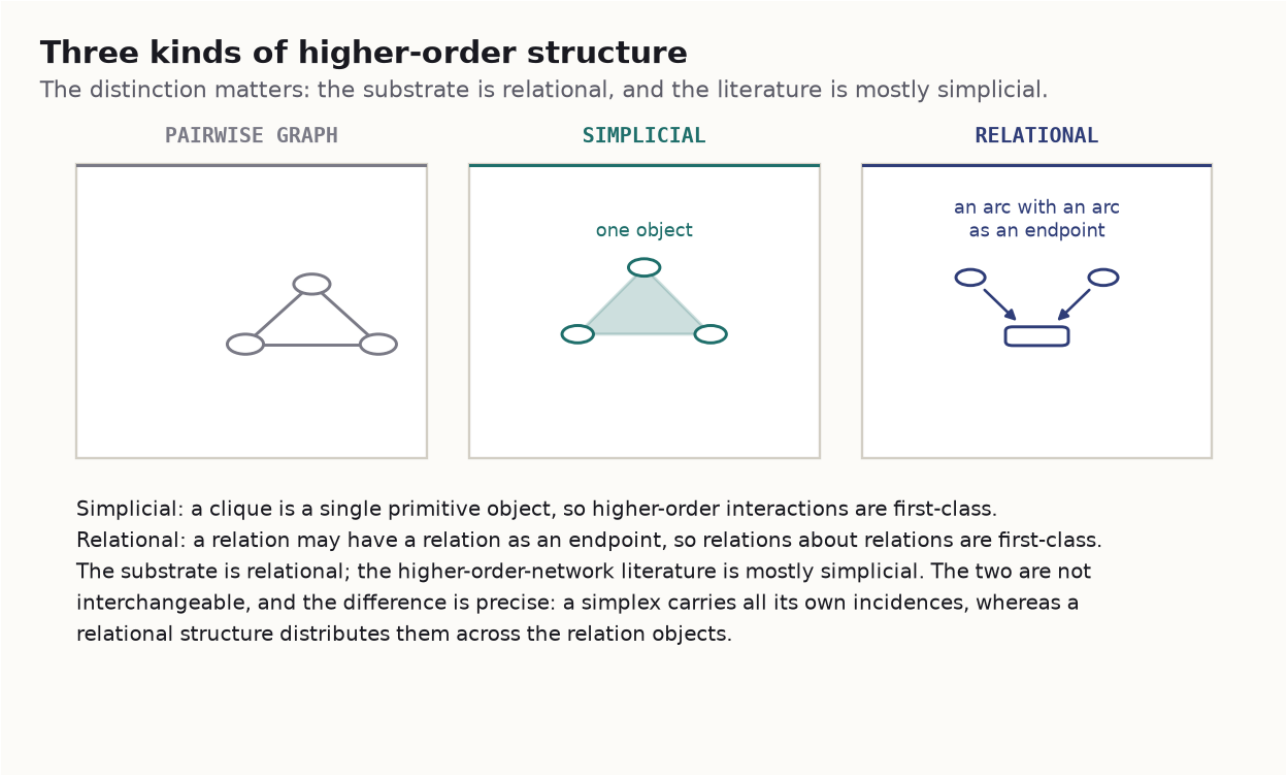


Figure 16. Three kinds of higher-order structure

19.1 The distinction is not cosmetic

The higher-order network literature has a standard move: take a clique of k nodes and treat it as a single primitive object with its own incidences. Battiston, Cencetti, Iacopini, Latora, Lucas, Patania, Young and Petri's *Physics Reports* review (2020) is the canonical statement, and Kivelä, Arenas, Barthelemy, Gleeson, Moreno and Porter's multilayer framework (2014) is the canonical formalism — in which a single pair of nodes may be connected in multiple layers with *different signs*, so that one layer can have positive or negative effective coupling.

The substrate is **not** simplicial. It is relational: a relation may have a relation as an endpoint, and the "higher-order object" is not a clique but the relation itself. The distinction is exact:

- A **simplex** carries all of its own incidences with it. Removing a vertex from a clique removes a vertex from the simplex.
- A **relational structure** distributes the incidences across the relation objects. A relation that has a relation as an endpoint does not contain it; it *refers* to it, and the referent can change.

This difference is not academic for the substrate, because it is what makes the "derived identity" commitment possible at all. In a simplicial model, a higher-order object is a thing; in the substrate, a higher-order object is a *pattern*, and patterns can be read differently at different times.

19.2 The theorem that makes the distinction testable

Benson, Gleich and Leskovec (2016), in *Science*, showed empirically that the higher-order (overlap) community structure of real networks is **not recoverable from the pairwise projection**. And Battiston and colleagues showed that dynamical phenomena exist in hypergraphs that have no pairwise counterpart: simplicial contagion, higher-order synchronisation, simplicial k-core percolation, and the emergence of inter-layer structures such as composite communities

and multilayer modularity.

This is the single most important result for the substrate's commitment 1, and it cuts both ways.

It gives the substrate a test it does not currently pass. The commitment "an arc may have an arc as an endpoint" is, until measured, a claim about architecture. The Benson–Gleich–Leskovec protocol asks whether the higher-order structure carries information that the pairwise projection does not — and the substrate's higher-order structure is *at the same time* its relation structure, so the projection question is subtle: what is the pairwise projection of a substrate? It is a graph on the same identifiers with arcs from arc-endpoints to node-endpoints, and the higher-order structure is the part that references arcs.

It also shows what the payoff would be. If the substrate's order-≥2 structure carries information beyond its dyadic projection, then the substrate has the higher-order analogue of a result with a peer-reviewed protocol behind it. If it does not, then commitment 1 is decoration, and the project should say so.

The relevant measurement protocol exists: higher-order structure scored against an ensemble matched to the relevant statistics, with the power reported alongside the z-score.

20. Topological signatures

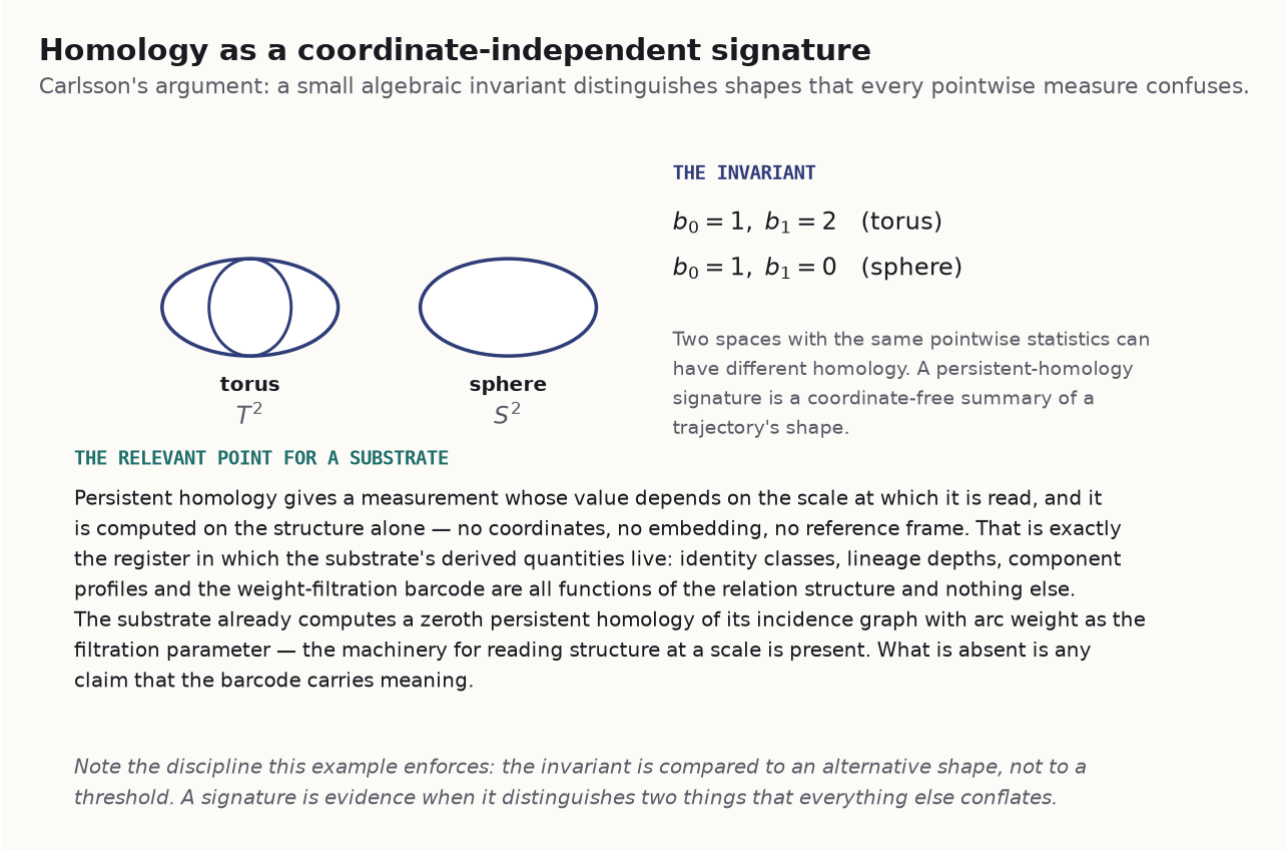


Figure 17. Homology as a coordinate-independent signature

20.1 The argument

Carlsson's argument is the cleanest statement of why topology is the right tool for a relational substrate. Homology assigns to a space a sequence of integers — the Betti numbers — that are *coordinate-independent*: they do not change under deformation. A torus and a sphere can have

identical pointwise statistics and different Betti numbers ($\beta_1 = 2$ against $\beta_1 = 0$). A signature is therefore evidence precisely when it distinguishes two things that everything else conflates.

For the substrate this is unusually apt, because the quantities of interest are already functions of the relation structure and nothing else:

- identity classes are connected components of the relation pattern spanned by the fusion witnesses;
- lineage depth is a longest-chain computation over the descent relation, with cycle detection;
- component profiles and degree distributions are structural statistics of the incidence graph.

And the substrate already computes a zeroth persistent homology with the arc weight as the filtration parameter. The machinery for reading structure at a scale is present.

20.2 What is missing, and it is not the machinery

What is missing is a **claim** that the barcode carries meaning, and the discipline that would support such a claim: a filtration chosen for a reason, a comparison with an alternative filtration, and a demonstration that the resulting signature separates structures that the substrate's other statistics do not.

The natural test follows the discipline the whole document has been insisting on: the signature should distinguish two things that everything else conflates. In the substrate's case, the candidate contrast is between a substrate whose rules were rewritten and one whose rules were not — a question about whether the *history* of a structure is visible in its topology, which is exactly what the tombstone set was designed to make visible.

20.3 The general relevance

Ghrist's result — that the Vietoris–Rips complex of a finite metric space captures its homotopy type — is the technical bridge between "a relation structure" and "a topological signature", and it is what makes the whole approach available rather than decorative. The Čech and Rips filtrations, and the theorem relating the stability of one to the other, are the formal apparatus; Ghrist's theorem is the statement that the apparatus applies to a finite relational space at all.

21. Dynamical systems and the geometry of regime change

The substrate's regime vocabulary — transient, fixed point, limit cycle, drift, chaotic, undetermined — is a coarse partition of a well-studied space. The three classical correspondences it contains are:

- **Fixed point.** Everything converges; the space is one point. The canonical elementary model is $\dot{x} = -x$.
- **Limit cycle.** A closed orbit; the canonical normal form is $\dot{r} = r(1 - r^2)$.
- **Riddled basins.** Milnor's 1999 construction showed that a chaotic attractor's basin structure can be infinitely fragmented, with the boundary of a basin having positive measure — so the attractor a trajectory converges to can depend on initial conditions in a way that no coarse classification captures. "Riddled basins" is the sharpest available warning against reading a regime label as a complete description.

Robust chaos (Garrigó and Sansone) sharpens the warning: some chaotic attractors are robust to parameter perturbation while others are not, so the question "is this trajectory chaotic?" has a

second, harder question attached — "is it robustly chaotic?" The substrate's classifier reports its measurements and has an explicit *undetermined* outcome for cases it cannot separate at the available checkpoint density, which is the right posture.

The computability boundary runs through this. A continuous map of the interval with a period-three orbit has orbits of every period and an uncountable scrambled set (Li and Yorke, 1975) — the sharpest elementary statement that local simplicity forces global complexity. And the impossibility results for cellular automata (§12.3) say that the long-term behaviour of a sufficiently general rule set cannot be decided in general. The substrate lives under both statements, and its one structural result is a statement about a limit cycle of a specific kind: a capacity-coupled closed loop in which one rule frees exactly the capacity another consumes.

What the substrate cannot do. It cannot certify that its rule set is non-settling in general, and §12.4 explains why: if the rule set is universal, that certification cannot be internal to it.

22. Information theory, complexity and the description gap

22.1 The bound that constrains everything

For any computable function f and any input x,

    $K(f(x)) \leq K(x) + K(f) + O(1)$

where K is prefix-free Kolmogorov complexity (Li and Vitányi, 1997; Chaitin, 1987).

The consequence, stated as it should be: **a program cannot produce an output whose algorithmic information content exceeds, by more than a constant, that of its own description together with its input.**

And the correction that matters is precise. The program does not need a description *of the structure*. It needs a description of *itself and its input*; every output is then fully, non-shortly described by (program, input, step index). What a program *can* produce is output with no **short** description: long, incompressible-looking behaviour that is nonetheless entirely computed.

That is the honest source of "the system generated something not designed". It is not new algorithmic information. It is a **description gap** between the system and any human summary of it.

22.2 The test this licenses

For the substrate, the bound becomes a measurement. Its global behaviour has $K \leq K(\text{rules + policy}) + K(\text{initial configuration}) + O(\log t)$. Any claim that the substrate "generated" structure not somehow already in its rules and initial state is false in the algorithmic-information sense — **and can be tested**: compute or bound the complexity of the substrate's macro-behaviour against the complexity of its micro-rules plus initial state.

If the macro-behaviour has a *much* shorter description, the substrate has produced a compression — a genuine emergent regularity — and that is the strongest available form of "generated something not designed".

This is the concrete form of the Israeli–Goldenfeld search of §5.3, and it is the sharpest single research direction the whole document produces.

22.3 Entropy, and the object the substrate actually has

Shannon's axioms make entropy a function of a distribution; Jaynes's maximum entropy principle then makes the distribution the primary object of inference rather than the configuration. The substrate has the distribution over realisations and does not have a distribution over *states* in any physical sense, nor an energy, nor a heat. The maximum-entropy treatment would apply to the ensemble of trajectories, which is exactly the object the null models sample — and the honest position, as §9 states, is that the second law's applicability to that ensemble is an empirical question, not a given.

The Gibbs variational principle is the formal frame in which such a question would be posed; and the substrate's absence of a generative model of itself means that the free-energy principle, as a *physical description*, would require smuggling in an agent that the substrate does not have.

23. Cellular automata, undecidability, and the ceiling

23.1 The taxonomy is not a theorem

Wolfram's four-class taxonomy — class 1 homogenises, class 2 stabilises into periodic structures, class 3 appears chaotic, class 4 produces complex localised structures that survive — is a heuristic classification, and Wolfram's own caveat is that any general scheme inevitably assigns borderline cases by judgement. Competing formal taxonomies exist. The classes are a useful heuristic and **not a mathematical object**, and the substrate should not be described as "class 4" as if that were a diagnosis.

Israeli and Goldenfeld's result (§5.3) substantially undercuts the popular reading that emergence is confined to class 4: coarse-grained descriptions can be constructed for automata in **all** classes. The taxonomy and the ceiling must be kept apart.

23.2 The ceiling, which is a theorem

Wolfram's 1985 result in *Physical Review Letters* is the peer-reviewed statement: physical processes can encode universal computation, and therefore there is no general algorithm that decides the outcome of such a process — a limitation on *prediction*, not merely on speed. Perales-Eceiza and colleagues' 2025 *Physics Reports* review gives the modern scope: undecidability has been established for phase transitions, thermodynamic quantities, spectral properties and quantum dynamics in various models.

The precise formulation the substrate must live under:

> If the substrate's rule set can simulate a universal computation, then no algorithm decides the substrate's long-term behaviour, and in particular there may be no general procedure that decides whether the substrate settles.

And the corollary that bears directly on the project's one structural finding: **the absence of the birth rule gives a fixed point, which is a decidable, checkable fact; the presence of it gives non-settling, which is potentially an undecidable one.** The two halves of the finding are not epistemically symmetric, and the project's own documents do not currently say so.

23.3 The completeness corollary

No consistent, effectively axiomatised formal system of complexity K can prove the halting of all programs of complexity below K (Chaitin). For the substrate the interpretation is sharp: **if its rule set is universal, then a proof that the substrate never settles cannot be internal to that rule set.** The structural finding would then be an empirical observation about an undecidable predicate — still worth having, but it must be described as such, and must not be presented as a theorem of the substrate's own logic.

This is why the two open experiments of §42.1 and §42.2 are the right ones to run first: *is the relation structure a partial order?* and *is the rule set universal?* Either answer strengthens the theory; a third answer — that the rule set is not universal and therefore a settling theorem exists — would strengthen it most.

24. Universal algebra, fixed points, and what Skolem costs

24.1 Fixed points

The substrate's engine halts when the possibility space is empty; its rules are monotone-ish rewriting operations on a partial order of states. The mathematical fact that governs this territory is Tarski's fixed-point theorem: a monotone map on a complete lattice has a least and a greatest fixed point. The Knaster–Tarski refinement gives the same for the iterative construction.

This is the formal shape of "a stable state exists": the substrate's fixed points are exactly what Tarski's theorem is about, and the fact that the engine finds one in finite time in the birth-absent case is a *decidability* result about a specific rule set rather than a general theorem.

24.2 Skolem, and the unbounded identifier space

The substrate's identifier space is unbounded in principle. The Löwenheim–Skolem theorem says that any theory in a countable language with an infinite model has a countable model; Skolem's 1928 construction gives it explicitly. The consequences for a substrate with an unbounded identifier space are real and mostly inconvenient: countable models of uncountable theories need not be isomorphic, so two substrates that agree on every finite fragment of the relation structure need not be isomorphic. That is the *Skolem paradox*, and it is a warning about a specific class of claim: that "the substrate's structure is determined by its local rules" is not automatically true in the presence of non-local identifications.

It is a reason to prefer *local* determinations — derivations that proceed by a bounded induction over the structure — over global ones. The substrate's derivations are local (the identity classes are connected components of a relation pattern; the lineage depths are longest chains), and that is the right design choice for a reason that is now nameable.

24.3 Homomorphism and what "structure-preserving" means here

A homomorphism between relational structures is a map preserving the relations. In the substrate, the relabelling used by the invariance battery is exactly a homomorphism from the state to its relabelled copy, and the structural digest is what makes the choice of labelling irrelevant. This is the precise formal content of the substrate's claim to be relabelling-equivariant, and it is why the *structural* digest and the *graph* digest are kept separate: the first is the canonical form of the

structure, the second is the provenance anchor of the particular world.

25. The mathematics the substrate has and the mathematics it lacks

Commitment of the substrate	The relevant mathematics	Status
Shared identifier space; relations about relations	Category theory; higher-order and simplicial network theory; hypergraph spectral theory	the family is right; the apparatus does not apply directly to a *dynamic* relational structure — `[OPEN]`
Tags as untyped integers; a reserved emergent range	Universal algebra; presheaf categories	the algebra of labels is trivially free; the *semantics* is deliberately absent, and that is the design, not an omission — `[IMPLEMENTED]`
Identity and lineage derived, never stored	Equivalence relations; connected components; fibre partitions of a map; neutral networks (Wagner 2008; Weiß and Ahnert 2020)	the derivation is implemented; the *prediction* about the structure of the derived classes is missing — `[HYPOTHESIS]`
No global clock; several local times coexist	Causal set theory; Markopoulou–Smolin causal order; partial orders	strong in form, fragile in hypothesis; a transitivity test would settle it — `[HYPOTHESIS]`
Local rules as data	Term-rewriting systems; Church–Rosser; Petri nets; Newman's lemma	the substrate's rules are rewriting operations over a labelled relational structure — `[CLASSIFIED: MATHEMATICAL]`
One site per step, by a stated policy	Stochastic processes; ergodic theory; Kuramoto-type self-consistency	the policy is a selection operator; an analogue of a critical coupling would be a real result — `[PROPOSED]`
Measurement against nulls	Statistical physics; the null-model critique of Girvan and Newman; Gelman and Stern on "not significant" against "not measurable"	implemented, and the discipline is an import rather than an invention — `[BACKED]`
Coarse-grained descriptions may exist	Israeli and Goldenfeld; the block encoding of Cook	the licence for the project's most ambitious experiment — `[OPEN]`
Retention of a tombstone for removed objects	Landauer is *inverted* here — the substrate retains what erasure would cost; this is bookkeeping, not thermodynamics	a category error to avoid — `[CLASSIFIED: SUGGESTIVE ONLY, misleading if used]`

Table 3.

Part V — Complex systems and the neighbouring disciplines

26. The complex-systems reading

26.1 What the substrate already is

Read through the complex-systems literature, the substrate is not an unusual object. It is a **population of heterogeneous individuals with a shared, sparse vocabulary and local interaction rules**, in which turnover and resource limits bound the population — and that is exactly the description of a dissipative structure, a traffic model, a sandpile, a developmental system or an autocatalytic network, depending on which formalism you bring.

What is unusual is the instrumentation, not the object. The complex-systems literature has produced a large number of *models* and a comparatively small number of *instruments for measuring models honestly*. The substrate is an instrument.

26.2 Three formalisms the substrate already satisfies

Replicator dynamics. For a population with n types, frequencies x_i with $\Sigma x_i = 1$ and payoff matrix A, the replicator equation is

$$\dot{x}_i = x_i (f_i(x) - \varphi(x)), \quad f_i(x) = (Ax)_i, \quad \varphi(x) = x \cdot f(x)$$

and it is dissipative: it preserves the simplex and possesses a Lyapunov function $V(x) = \Sigma x_i \ln x_i$ along solutions, which guarantees a compact invariant set and restricts the dynamics to fixed points, heteroclinic cycles and — nongenerically — periodic orbits.

The substrate's engine is not a replicator dynamic, and should not be described as one. But the *structural parallel* is instructive: the substrate has a population with types (the tags), a shared state space (the substrate), and bounded resources (the node and arc caps), and it too has a Lyapunov-flavoured question — what quantity decreases?

Spatial games. Nowak and May's 1992 spatial Prisoner's Dilemma is a *purely deterministic* system with no memory among players and no strategic elaboration, and it generates "*chaotically changing spatial patterns, in which cooperators and defectors both persist indefinitely*". Lieberman, Hauert and Nowak's generalisation replaces the lattice with an arbitrary graph, making the *structure of the interaction network* a determinant of the evolutionary outcome rather than a background condition.

The substrate is the same shape: a deterministic local-rule system on a relation graph, whose *structure* is the thing that varies. The substrate's structural campaign (§7.4) is a Nowak–May-shaped experiment in a different variable.

Adaptive networks. Gross and Blasius's review is the formal statement of fitness-based rewiring: each node has a fitness $f_i = \Pi_{k \in N(i)} F(a_k, b_k)$, with the usual choice $F(a,b) = e^{\beta * a * + b}$, and the probability that a rewiring event targets node j is proportional to f_j. *The review's contribution is the* taxonomy* and the demonstration that fitness-dependent rewiring produces stationary distributions that are genuinely different, and often non-ergodic.

The substrate's natural counterpart is its own policy choice. Which rule fires is decided by a policy, not by the rule set alone. The substrate therefore has an *adaptive* dimension already, and it has never been explored as one. What would it mean for the selection policy itself to be a function of the state? That is a well-posed experiment with an established formalism, and the substrate is already instrumented to run it.

26.3 A clean micro-to-macro precedent

Nagel and Paczuski's emergent traffic jams (1995) is one of the better-documented cases of a microscopic model with **no tuned parameter** producing a macroscopic critical state: the outflow from a jam self-organises to a critical state of maximum throughput, small downstream perturbations create jams with a power-law lifetime distribution $P(t) \sim t^{-3/2}$, and on varying vehicle density the critical state separates lamellar and jammed regimes with $1/f$ noise.

The substrate has the same ingredients and no equivalent result: a microscopic local rule, no tuned parameter, a macroscopic critical state separating two regimes, and a power-law distributional signature waiting to be measured.

26.4 The landscape

Wright's adaptive landscape is a map from genotypic space to fitness, where the fitness of a genotype depends on the genotypes it can be reached from, and epistasis creates local optima separated by valleys. The landscape is *context-dependent*: the same genotype has different fitness in different backgrounds.

The substrate's structural space — rule subsets × scheduler policies — is a landscape in exactly this sense, and the substrate has never been described as one. The nearest analogy is a *neutral network* landscape, and §30.2 makes that concrete.

27. Theoretical biology and the origin of life

27.1 The resemblance, and its exact size

The substrate is, in the biological register, a **population of heterogeneous individuals with local catalytic interactions, bounded by a resource ceiling, in which individuals are born, merge and die**. That is the Kauffman picture of the origin of life, and the formal apparatus is the autocatalytic set.

An **autocatalytic set** is a set of catalysts each of which is catalysed by another element of the set; it is **irreducible** if no proper subset is autocatalytic. Hordijk and Steel's framework makes detection algorithmic and *decidable*: the catalyst network is a directed graph, and finding a self-sustaining set is a graph-theoretic problem.

Two of its structural notions bear directly on the substrate.

Enablement — a catalyst in the set that is *produced* by the set but not currently *used* by it — measures the latent capacity of the set to grow.

Evolvability — how many new catalysts can be added while keeping the set self-sustaining — is the size of the *adjacent possible* of an autocatalytic set, and Kauffman's adjacent possible and Hordijk–Steel's evolvability are the same quantity computed two ways.

The key fact for this document: **evolvability is a global property of the catalytic network, not a property of any local node.** It is computed from the network as a whole and is not determined by any individual's behaviour.

The substrate has the same structure. Its fusion statistic — the number of identity classes of size greater than one, and the fraction of members in them — is a global property of the relation pattern, not of any object. Its lineage depth is likewise global. Whether the substrate's derived identity classes have an *evolvability*-like structure — a set of latent configurations reachable by one step — is an open question with a precise formal analogue in the biological literature, and it is the strongest available way to make the substrate's "derived identity" commitment into a model with predictions rather than a definition.

27.2 The thermodynamic requirement the substrate does not meet

Life as a dissipative structure requires an open system sustaining a stationary non-equilibrium state by exporting entropy faster than it is produced: $dS = d_{\mathrm{e}}S + d_{\mathrm{i}}S$ with $d_{\mathrm{e}}S < 0$ balancing $d_{\mathrm{i}}S > 0$. Kauffman and Roli's recent treatment argues the emergence of life — and of agency — as an *expected* transition with a computable criterion rather than a miracle.

The substrate has no energy and no entropy. It is not a dissipative structure and should never be described as one (§6.1). What it can legitimately offer the origin-of-life programme is much more modest and more interesting: **a computational laboratory for the combinatorial part of the question** — which autocatalytic sets exist in a catalytic network of a given size, which are irreducible, which are enabled, and how those counts scale — with a decisive detection procedure and a decidability result already in the literature.

27.3 The restricted claim that is legitimate

There is one claim the substrate can support about biology, and it should be stated narrowly:

> The substrate is a laboratory for **population dynamics of local-interaction systems under resource constraints**, in which individual lifetimes, turnover, population size and the structure of merging are observed rather than assumed, and in which identity and lineage are read off the interaction pattern instead of being declared.

The questions that setting makes answerable are the demographic ones: what is the distribution of individual lifetimes; how does the population size behave as the resource ceiling is varied; is there a critical ceiling at which the population's *structure* changes qualitatively?

They are not the questions of molecular biology. The substrate has one level of organisation and no interior, no metabolism, no genome and no developmental programme; a cell is organised by an internal network, and the interesting dynamics in biology are often the interaction *between* levels of organisation. The substrate has exactly one level, so it cannot exhibit the nested organisation that makes biology distinctive. §20's higher-order network result (Benson, Gleich and Leskovec) is a warning in the same direction: a higher-order representation that adds nothing beyond its pairwise projection is decoration.

28. Information sciences and the theory of observation

The substrate's most mature intellectual contribution is not a result but a distinction, and the distinction has a name in statistics and in the information sciences.

Gelman and Stern's *Difference between "significant" and "not significant" is not merely a matter of usage* exists to say that a measurement which failed to reject and a measurement which could not be made are different claims, with different implications for what should be believed. The substrate's entire verdict vocabulary is built around that distinction, and the higher-order network literature has carried the same discipline into exactly the ensembles the substrate would need.

Three observations from the information sciences bear on the substrate's architecture.

Observability is a choice, and it should be declared. The substrate has fifty-three observables and synthesises thirty more, but the project admits it cannot yet ask for the raw bank versus a selected subset. Any measurement regime is a choice about what counts as evidence, and the choice has to be part of the record.

Description length is a currency. The Kolmogorov-complexity bound (§22.1) turns "the system found something I did not put there" into a measurement: compare the complexity of the macro-behaviour with the complexity of the micro-rules plus initial state.

Replication is a discipline, not a courtesy. The project's own confirmed result includes a rival hypothesis that was falsified 0/24 and another left inconclusive 11/24, and the mechanism's confirmation rests on twenty-four held-out seeds. That is the practice the substrate is built to make possible.

29. Theoretical computer science: the substrate in formal terms

The substrate is, in computer-science terms, a **labelled transition system over a relational state space, driven by a term-rewriting system, with a non-deterministic site-selection step and a decidable-enough analysis layer**. Setting out the correspondence precisely matters, because it identifies which results transfer and which do not.

Labelled transition systems. A state, a set of labelled transitions, and the induced reachability relation. The substrate's states, its rules and its possible sites form exactly this. The relevant classical results — bisimulation and behavioural equivalence, the theory of nondeterministic branching semantics, the superposition and rewriting tradition — are the right literature for questions about what can be observed of a transition system.

Term rewriting. A rewriting system is a set of rules over an alphabet, where rewriting replaces occurrences of a left-hand pattern by a right-hand replacement. The substrate's rules are *contextual* rather than syntactic: the left-hand side is a predicate over bounded neighbourhoods, not a pattern. The relevant property is the **Church–Rosser property** — local confluence plus termination gives confluence — and the relevant theorem is **Newman's lemma**. The substrate does not claim confluence; its rules are deliberately non-confluent, and the site-selection policy is the tie-break. What would be worth testing is whether the *observable* behaviour is confluent in the sense that matters: whether different selection orders produce the same *statistics*. The project's regime classifier and trajectory comparison are the instruments for that, and the question has never been posed in this form.

Petri nets. Places, transitions, tokens, firing rules, and the reachability problem — which is undecidable. The substrate's rules that consume capacity and release it are, structurally, net-like; and the fact that Petri net reachability is undecidable is a *warning* about the substrate's own central question: whether the substrate settles is a reachability question in a system of exactly this kind.

Universal computation in continuous and discrete systems. Moore's generalised shifts and the Blum–Shub–Smale model formalise computation in continuous spaces, and the substrate's finiteness and discreteness put it on the other side of that divide — but nothing prevents it from *simulating* a universal machine, and §23.2 and §42.2 explain why that single fact would determine the epistemic status of the project's one structural result.

What the substrate is in one line. A finite, deterministic, relational, locally-rewritten state space with an explicitly enumerated possibility space, a declared null model per measurement, and a one-verdict resolution procedure. That is an unusual object. It is not an unusual *kind* of object, and the literature on all of the above already exists.

30. AI and scientific discovery

30.1 The demonstrated systems

The systems that have actually demonstrated machine-assisted discovery are worth naming precisely, because the substrate is closer to them than is usually admitted.

FunSearch (Romera-Paredes and colleagues, *Nature* 625, 468–475, 2024) is the structural template: a large language model proposes code; an automated evaluator filters it; the survivors are used to improve the proposer. The result was a discovery in the cap-set problem that improved on the best known human construction — and, importantly, the discovery was made by the *evaluator*, not by the model.

AlphaProof and the AI co-scientist line (Sakana AI, 2025) extend the pattern towards theorem proving and hypothesis generation, with the same separation between proposal and validation.

The Darwin Gödel Machine (arXiv:2505.22954) is the reference for self-modification in a sandbox, with evaluation as the filter.

SINDy (Brunton, Brunton, Proctor and Kutz, *SIAM Journal on Applied Dynamical Systems* 20(4), 2154–2182, 2021) is the reference for *not assuming the terms*: regress a library of candidate terms against measured derivatives and keep the few that matter. The substrate's representation synthesis is the same move with a different object — candidate *statistics* rather than candidate *terms*.

30.2 The substrate's actual position: the validating half already exists

What a machine-assisted discovery loop needs is: a space of hypotheses, a proposal mechanism, an evaluator, a multiplicity control, a confirmation stage, and a durable record.

The substrate has five of the six, and they are the hard ones:

Need	The substrate
A space of hypotheses	the possibility space of sites; the structural space of rule subsets × policies
A proposal mechanism	**absent** — and its absence is a design decision, not an omission
An evaluator	five gates, one verdict resolver, a diagnosis layer
Multiplicity control	registered protocols with content hashes; the campaign that reported UNDERPOWERED rather than a false negative
Confirmation	independent seeds derived from a protocol offset; a rival falsified 0/24 and a rival left inconclusive 11/24
A durable record	a store with four replay guarantees and divergence attribution

Table 4.

The validating half is the part that must not be delegated, and the project says so: an LLM wrapper would destroy the only property that makes its claims credible.

30.3 The failure mode, quantified

A human proposes a handful of candidates; an automated proposer proposes thousands. The correction for m tests must be expressible at the alpha level used, which requires a null budget of order $1/\alpha$ per test — m/α surrogate draws in total. The project's own campaigns were UNDERPOWERED at 250–400 surrogates over 191 tests, and said so.

The multiplicity problem is the single most likely failure mode of AI-assisted discovery, and it is the reason the substrate's power accounting has to be part of the proposal interface and not merely part of the report. A system that lets a model propose without charging for the multiplicity of the proposals will produce confident nonsense at a scale no human reviewer can audit — and the substrate's own gates, which are entirely capable of refusing such a result, are exactly the mechanism that would catch it.

30.4 What the substrate would contribute to the AI-for-science programme

Not a proposer. A **certifier**: the capability to take a proposed statistic, a proposed protocol and a proposed claim, and return — deterministically — whether the claim is supported, refuted, or not measurable, with the reason and the power analysis attached. That is a component an AI-driven discovery system needs and that no current system provides, because the validation layer of every existing system is either human or the same model that proposed.

31. Cosmology and the foundations of physics

The one cosmological result that bears directly on the substrate's commitments is the **problem of time**, treated in §16. The others are worth stating in the negative.

The problem of time is the only place where the substrate's *refusal* is the same shape as a frontier problem in fundamental physics: the Wheeler–DeWitt constraint removes the external parameter, and the literature's consensus is that time must emerge from something. The substrate refuses to impose a global clock and has several local times instead — a posture that resembles the problem rather than solving it, and §16.4 gives the test that would turn the resemblance into a claim.

Decoherence and einselection would be the natural candidate for an account of how a substrate's global structure could be *selected* rather than merely *measured* — its selection of one site per step among many eligible ones is, structurally, a symmetry-breaking-like selection, and environment-induced superselection is the physics of exactly that. But the substrate has no environment, no entanglement, no decoherence, and no physical realisation; the analogy is that of a formal resemblance, and the *interesting* part of it is that it suggests a design question — *what would play the role of the environment in a substrate whose medium is its own history?* — rather than a physical claim.

Cosmological structure formation — the growth of structure from initial fluctuations — is the closest observational analogue to the substrate's birth/fusion balance, and the analogy is instructive precisely because it fails in a specific way: cosmological structure formation happens in a *dynamical* arena (general relativity) with a conserved quantity (energy-momentum) that bounds it, and the substrate's capacity bound is a hard cap imposed by the configuration, not a conserved quantity emerging from the dynamics. The difference is not cosmetic, and the substrate's fixed-point/non-settling result should not be read as a model of structure formation.

32. The connections that do not hold

A research document that lists only successes misrepresents the state of the art. These are the connections that look appealing and are wrong, or at least unsupported.

Landauer applied to the tombstone set — a category error, and in the wrong direction. Landauer's principle says that erasing one bit costs at least k_B ln 2, *conditional* on logical irreversibility and equilibrium memory. The substrate's tombstone set deliberately **retains** the identifier. So the substrate is, if anything, *anti*-Landauer: it keeps exactly the information whose erasure would be thermodynamically costly. The popular move — "our tombstones are Landauer's principle at work" — is wrong twice: there is no thermodynamic cost or benefit in a computational substrate unless it is physically instantiated, and the direction of the effect is inverted. The tombstone is a *bookkeeping* device; it is not an erasure, and it is not thermodynamics. `[CLASSIFIED: SUGGESTIVE ONLY, and misleading if used]`

The free energy principle as a description of the substrate. The free-energy principle is defined over a *generative model* of the system and its environment; states carry a distribution, and updates minimise variational free energy. The substrate has no generative model of itself, no inference procedure, no preference ordering over states, and no agent. Mapping it onto the substrate requires smuggling in an agent the substrate does not have. And the principle's own standing is contested: it has been argued to be a formal framework rather than a physical theory with independent empirical content, and its use as a generative theory has been challenged. Using it as a *design heuristic* is defensible; using it as a *physical description* is not. `[CLASSIFIED: SUGGESTIVE ONLY as physics; METHODOLOGICAL at best as a heuristic]`

Entropic gravity as an explanation of the substrate's growth. Every premise — a horizon, a Hilbert space, an energy with a thermodynamic distribution, a large-N limit — is absent. The *shape* of the argument (information content → apparent geometry) is present on both sides, but the inference does not survive the transfer. `[CLASSIFIED: SUGGESTIVE ONLY]`

"Invariance of a statistic implies conservation of a quantity" — there is no such general theorem, and Noether's theorem does not say it. See §8.2. `[CLASSIFIED: the general claim is UNSUPPORTED; the method is METHODOLOGICAL only]`

Wolfram's class-4 taxonomy as a diagnosis. The taxonomy is a heuristic, not a theorem, and Israeli and Goldenfeld construct coarse-grained descriptions for automata in all classes. Do not describe the substrate as class 4 and do not cite the taxonomy as a theorem. `[CLASSIFIED: SUGGESTIVE ONLY for the taxonomy; MATHEMATICAL for the ceiling]`

"Order from disorder" in the Kantian sense. The Kantian usage is teleological; the physicists' usage is a different claim with a different logical structure, and the two are not interchangeable. The nearest verified anchors are the self-organised-criticality and Anderson's spin-glass papers, and the phrase licenses no inference here. `[CLASSIFIED: SUGGESTIVE ONLY]`

"Symmetry breaking as a loss of information." The intuition is standard — a symmetry is broken by the system selecting a branch, a reduction in the number of distinguishable states — but the precise statement would need to say *what* information is lost and in *which sense*, and no citable primary source stating it in that form was located. This is a declared gap, not a claim. It is also, for exactly that reason, the most promising unexploited target for the substrate's own theory: the substrate's "one eligible site per step, chosen by a policy" is a symmetry-breaking-like selection among alternatives, and the substrate could *derive* the information content of that selection.

An asymmetry that must be preserved. In holography, the inference from information to geometry is a conjectural identification; in causal sets and in network geometry, the inference from relations to geometry is a theorem. Any document that cites AdS/CFT for the first and causal sets

for the second must keep the two epistemic grades apart. The substrate belongs, if anywhere, on the second side.

A triple confusion to avoid. The substrate's "no global clock" is not relativity's "no preferred frame" — relativity has a global time function in each inertial frame and a light cone — and it is not causal sets' "no global time function", which follows from a locally finite poset being non-totally ordered. The substrate may have neither. `[CLASSIFIED: STRUCTURAL at best, contingent on a transitivity test that has not been run]`

Part VI — The connections that are not obvious

This part consolidates and extends the connections that are worth a researcher's attention. Each is stated as a connection, classified by its kind, given the shared formal object, given what could be investigated, and given what would falsify it. The connections that fail are in §32 and are not repeated here.

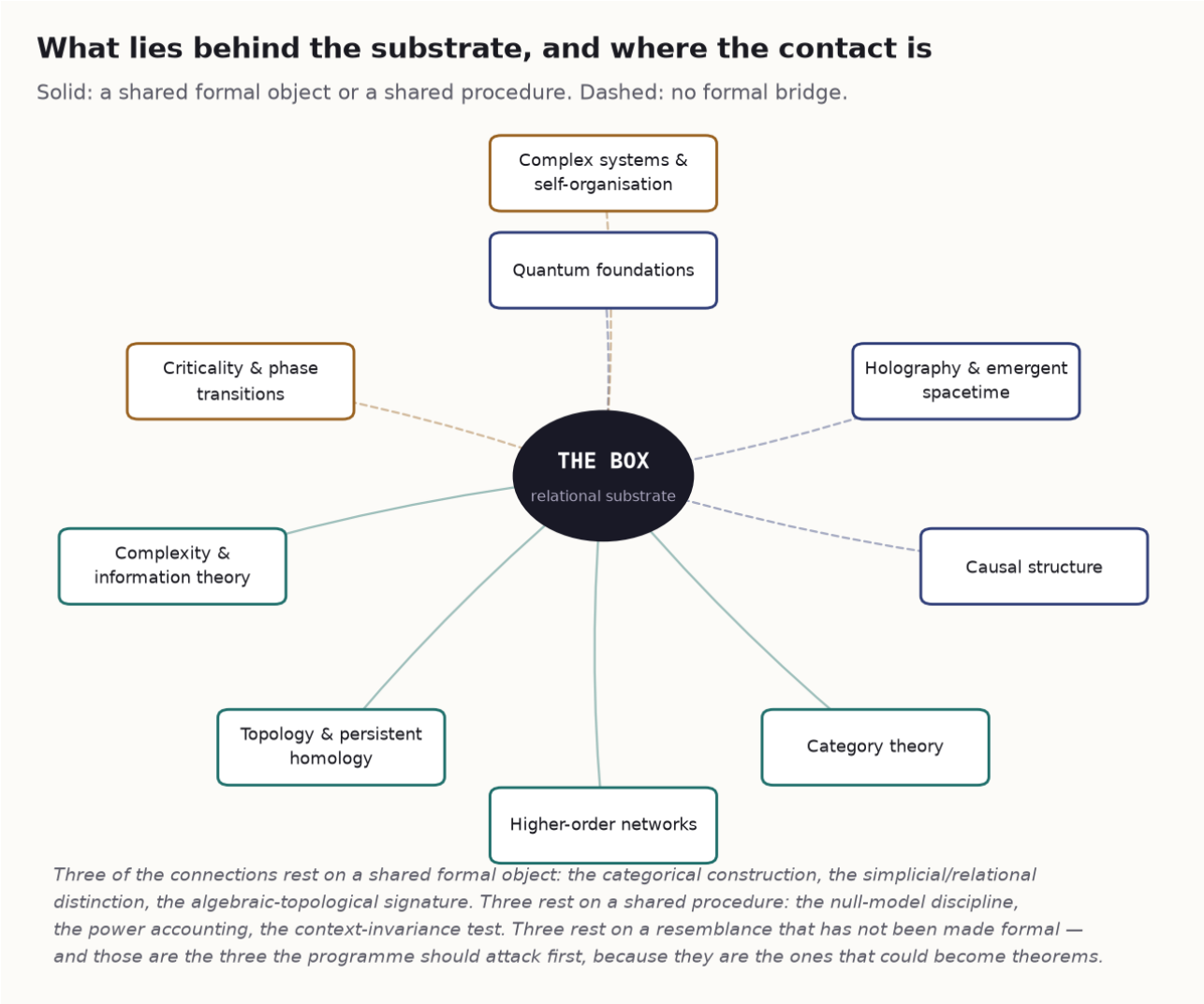


Figure 18. What lies behind the substrate

The map's two line styles are the whole of the calibration: a solid line means a shared formal object or a shared procedure exists; a dashed line means the resemblance has not been made formal. Three connections are solid, three are methodological, and three are dashed — and the three dashed ones are, deliberately, the ones the programme should attack first, because they are the

ones that could become theorems.

33. The strongest: coarse-grained descriptions exist even for undecidable systems

Israeli and Goldenfeld (2006) is the single best find, and it does three things at once that the substrate cares about.

It removes a false restriction. Coarse-grained descriptions can be constructed for cellular automata in *all* behavioural classes — so "complexity is at the edge of chaos" is not a boundary condition on where emergence can happen. The popular taxonomy of Wolfram is not a theorem, and it is not a boundary.

It puts emergence and undecidability in the same room. Some of the coarse-grainable automata are universal Turing machines and therefore undecidable. So the substrate's central ambiguity — *did the birth/fusion balance produce the fixed point, or is the balance itself an undecidable property?* — is a question the literature has already shown to be of the right kind, rather than a sign of a badly posed experiment.

It supplies a theorem-backed version of "generated something not designed". The probability of finding a coarse-grained description approaches unity as the scale coarsens, because the large-scale rule is often much simpler than the primitive one. Their conclusion is that large-scale patterns are "*forced upon the system by the simplicity of the rules that govern the large-scale dynamics.*"

`[CLASSIFIED: MATHEMATICAL]`

What it licenses for the substrate. The claim that the birth/fusion balance produces a fixed point in one case and non-settling in the other is a claim about the existence of an *effective description at a coarse scale*. The right experiment is therefore to search for an effective rule set of lower descriptive complexity that reproduces the large-scale behaviour — and to measure the complexity of the candidate effective rules against the primitive rules, rather than judging "simpler" by eye. This is the sharpest research direction the whole investigation produces, and it is §42.3.

What would falsify it. If no effective rule set of lower complexity reproduces the large-scale behaviour, then the substrate's macroscopic description is not a coarse-graining of its microscopic one, and the project is measuring a regime rather than a law.

34. The strongest formal object for derived identity: the fibre partition of a map

The substrate's third commitment — identity classes and lineage depths are *derived* by reading the pattern of relations, never stored — is precisely the **fibre structure of a map**, with the *neutral components* of the map as the connected classes inside each fibre.

Wagner (2008) established the map and the neutral-network structure: a genotype–phenotype map is a continuous function on a high-dimensional genotype space, and the connected components of its level sets are neutral networks — sets of genotypes with identical phenotype. Weiß and Ahnert (2020) then showed that the neutral components in a real genotype–phenotype map (RNA secondary structure) carry a **hierarchical community structure**, and that this structure is *not* recoverable from the pairwise projection of the map.

`[CLASSIFIED: MATHEMATICAL on the formal-object claim; the transfer is the author's own construction, not a literature statement]`

What this gives the substrate that it does not have. A falsifiable prediction about the structure of its own derived identity classes: if the classes are fibres of a well-defined map, then the neutral components inside each fibre should exhibit hierarchical community structure — and the measurement protocol for that already exists in the RNA-folding literature.

Three tests, in order of value.

1. Construct the map whose fibres are the identity classes. (Does such a map exist? The classes are equivalence classes of a relation pattern; the question is whether the pattern is the level set of a function on a suitable space.)
2. Test whether the neutral components inside each fibre form a hierarchical community structure.
3. Test whether lineage depth corresponds to a computable function of the map.

What would falsify it. If the derived classes are not fibres of any well-defined map — if the pattern-derived classes are not level sets of anything — then neither the causal-set nor the genotype–phenotype reading applies, and commitment 3 remains a definition rather than a model.

35. The strongest formal claim: a metric from a partial order

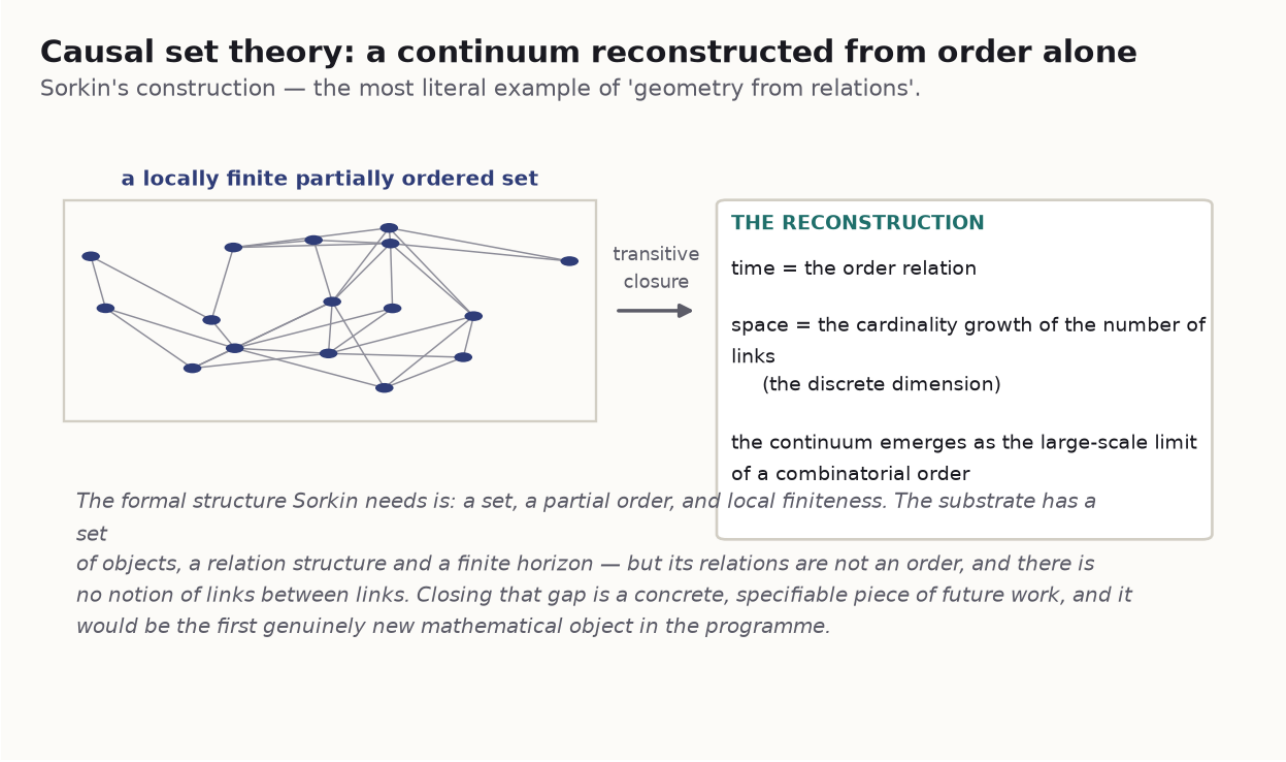


Figure 19. Causal set theory

Causal set theory is the rare case where "geometry from relations" is a theorem rather than a conjecture. Bombelli, Lee, Meyer and Sorkin (1987) showed that a locally finite partially ordered set with a measure supports a continuum Lorentzian geometry; Rideout and Sorkin (2001) recovered the continuum by a sprinkling and renormalisation argument; Dowker, Imambaccus, Owens, Sorkin and Zalel (2020) showed how the emergent metric is fixed up to a global conformal

factor and how the dynamics respect the emergent causal structure.

`[CLASSIFIED: MATHEMATICAL]`

The shared formal object is a locally finite partial order on a relational structure, together with the *order geometry* construction: spacing between two elements defined by the cardinality of their common past.

The catch, and it is a real one. The causal-set machinery requires **transitivity**, **local finiteness** and a **measure**. The substrate's "several local times coexist" is exactly the condition under which that machinery would fail — unless the times can be made consistent into a partial order.

What would falsify the connection. Two things, and both are cheap to check. If the substrate's local times are *revisable* — a later step can change an earlier relation's time — then there is no order to derive and the construction does not apply. And if the time on a relation is a *parameter of the rule* rather than a property of the relation pattern, then it is a timestamp and the parallel to relational time is superficial.

36. The strongest resonance in physics: derived time, not assumed time

The literature on the problem of time is largely a statement that time is *absent*. The papers that **derive** it are less cited and are the ones that matter here.

Kauffman and Smolin, in *Towards Quantum Gravity*, and then Markopoulou and Smolin (2004), define the causal order on a relational structure of states by the algebra of local transition operators, and derive locality intrinsically. Quantum Graphity (Konopka, Markopoulou and Severini, 2008) starts from a discrete state on a graph, evolves it with local rules only, and derives both the continuous spacetime *and its locality*; Hamma, Markopoulou, Lloyd, Caravelli, Severini and Markström (2010) give an exactly solvable variant in which the evolving graph supplies the emergent geometry.

`[CLASSIFIED: MATHEMATICAL]`

The shared formal object is a relation space on which a global order is defined by an algebra over local structure, with no background geometry assumed — and the substrate makes the same move for both time-like ordering and identity, from one relation space.

What the physics papers have that the substrate does not. Their order is a genuine partial order: it is *intrinsic* to the structure, and its consistency is part of the construction. The substrate's several local times have no stated consistency. **The difference is the whole content of the connection** — and it is testable, as §16.4 sets out.

Two directions exist and they are opposite. Markopoulou's later argument is that space can be emergent while time cannot — space does not exist, so time can — which is the mirror image of the substrate's own asymmetry, where time is refused and space is never attempted. Two systems with opposite asymmetries are not the same system, and the comparison is informative rather than flattering.

37. Two routes to a metric that are not holographic

Bianconi and Rahmede (2017) derive hyperbolic network geometry from relational data plus an ensemble; Millán-Torres, Torres and Bianconi (2019) read a spectral dimension off a relational structure.

These are not the AdS/CFT story and they require no quantum-gravity interpretation. For a computational substrate they are far more immediately usable than holography, because they require only three things the substrate already has: relational data, an ensemble, and a functional of the structure.

The physics of holography gives the ambition; these give the toolbox. And there is a third route the causal-set papers suggest, which the substrate has not tried: define a **spacing** from the size of relational neighbourhoods, test whether it satisfies a triangle-inequality-like constraint, and check whether the recovered dimension is stable across scales. If it is, the substrate has a metric derived from relations in the exact sense of Sorkin — and the holographic literature becomes an ambition rather than a necessity.

17.1 A gap that is genuinely open, and worth naming

Three literatures use hypergraph-like structures — network science's higher-order motifs, the contextuality-scenario hypergraphs of the quantum-foundations literature, and the quantum-information-theoretic "higher-order entanglement" of hypergraph quantum states and random tensor networks — and they **share a combinatorial vocabulary without sharing results**. No peer-reviewed work was found that takes a network-theoretic higher-order statistic and shows that it controls, or is controlled by, an entanglement or contextuality measure.

The reason is visible once the three are set side by side. In the network-science case the hypergraph encodes *interactions between agents*; in the contextuality case it encodes *a set of measurement procedures*; in the quantum-information case it encodes *a many-body wavefunction*. The combinatorial object has the same shape in all three and the semantics are entirely different.

This is a real gap, and it is the kind of gap a substrate whose relations may be relations is unusually well placed to occupy. The substrate's hypergraph-shaped structure is the *dynamical substrate itself*, not an encoding of something else. Whether that is an advantage depends on exactly the experiments of §45.5 and H5 — and it is a gap, not a result.

38. The substrate's own finding may sit inside the undecidability result it is trying to escape

This one was not sought; it is a consequence of putting the two literatures side by side.

Wolfram's 1985 result and the 2025 *Physics Reports* review establish that a system encoding universal computation has outcome predicates that no algorithm can decide. If the substrate's rule set can simulate a universal computation — which is exactly what "never reliably settles" would suggest if the birth/fusion balance creates an unbounded recursion — then the substrate's structural finding is not merely a structural fact but a **provably undecidable** one, and no amount of further preregistration will settle it.

The practical consequence is a programme, not a conclusion: determine whether the rule set is universal, by constructing an explicit embedding of a universal computation into the birth-and-fusion machinery. Because that single fact determines whether the project's one structural result is an empirical discovery about the substrate or a restatement of the halting

problem.

And there is an asymmetry worth exploiting. The *absence* of the birth rule gives a fixed point — a decidable, checkable fact. The *presence* of it gives non-settling — potentially an undecidable one. The two halves of the finding are not epistemically symmetric, and the project's own documents do not currently say so.

39. A conservation law that exists only at the coarse-grained level

The sharpest negative result in the survey is that there is no general theorem of the form "invariance of a statistic under a transformation implies conservation of a quantity" (§8.2). But there is a closely related *existential* result, and it is the one the substrate's own architecture resembles.

Baez, Pollard, Lorand and Sarazola have shown that **conservation laws can hold at the coarse-grained level and fail at the level of the individual components** — in a categorified treatment of biochemical reaction networks, where the emergent object is the morphism structure rather than the underlying set.

`[CLASSIFIED: MATHEMATICAL, but existential rather than general]`

This is the precise shape of what the substrate could look for: a quantity conserved under the *coarse-grained* description of its rule set but not under the primitive description. The project's trajectory comparison and regime machinery are the instruments for finding one; the categorical result is the precedent for the shape of the finding and the reason the search is worth doing.

40. The computability ceiling, stated exactly

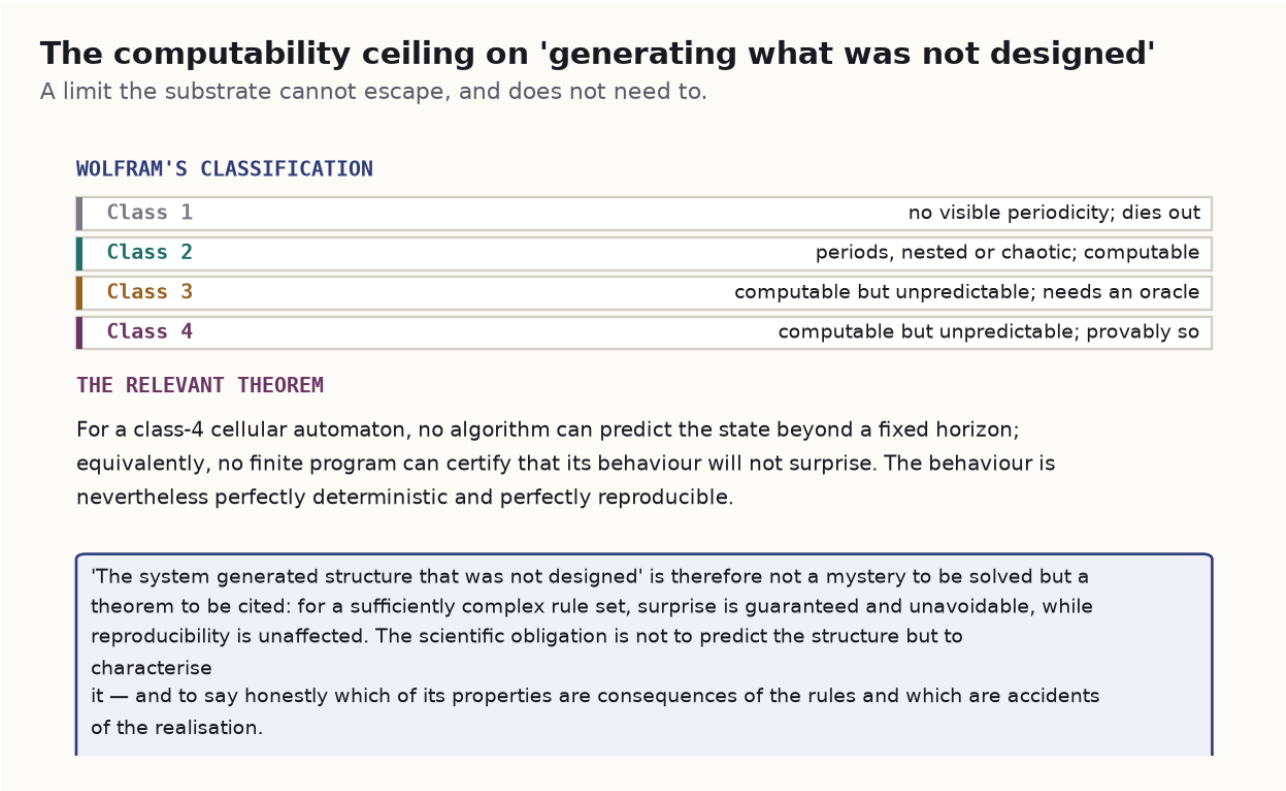


Figure 20. The computability ceiling

40.1 The bound

For any computable function f and any input x:

    $K(f(x)) \leq K(x) + K(f) + O(1)$

where K is prefix-free Kolmogorov complexity (Li and Vitányi, 1997; Chaitin, 1987).

The correction to the popular statement is precise and important: the program does not need a description *of the structure* — it needs a description of *itself and its input*. Every output is then fully, non-shortly described by (program, input, step index). There is no output without a description. What a program *can* produce is output that has no **short** description: long, incompressible-looking behaviour that is nonetheless entirely computed.

That is the honest source of "the system generated something not designed": not new algorithmic information, but a *description gap* between the system and any human summary of it.

40.2 The corollary for the substrate

The substrate's global behaviour has $K \leq K(\text{rules + policy}) + K(\text{initial configuration}) + O(\log t)$. Any claim that the substrate generated structure not somehow already in its rules and initial state is false in the algorithmic-information sense — **and it can be tested**: compute or bound the complexity of the substrate's macro-behaviour against the complexity of its micro-rules plus initial state.

If the macro-behaviour has a *much* shorter description, the substrate has produced a compression — a genuine emergent regularity — and that is the strongest available form of "generated something not designed".

40.3 The ceiling proper

If the substrate's rule set can simulate a universal computation, then no algorithm decides its long-term behaviour, and in particular there may be no general procedure that decides whether the substrate settles. The completeness corollary sharpens it: **if the substrate's rule set is universal, then a proof that the substrate never settles cannot be internal to that rule set.**

40.4 What the substrate genuinely can generate "not by design"

1. **A coarse-grained description that was not authored** — Israeli and Goldenfeld's result. This is the only rigorous available form, and it is the template the substrate should be measured against.
2. **Behaviour that no short program predicts but that the rule set computes.**
3. **Behavioural diversity discovered by search** — novelty search finds behaviour not encoded in an objective. But note carefully: novelty search explores behavioural diversity *within a fixed representation and a fixed set of primitives*. It does not create algorithmic information and it does not create a new vocabulary. It is the substrate's commitment 2 made operational, and it is a search procedure, not a source of new descriptions.

40.5 The honest one-line statement of the ceiling

> A computational substrate can produce behaviour that is undescribed, uncompressed and unpredictable; it cannot produce algorithmic information it did not already contain in the description of its rules and its initial condition; and if its rules are universal, it cannot prove anything about its own settling behaviour.

41. The two ladders, and the gap between them

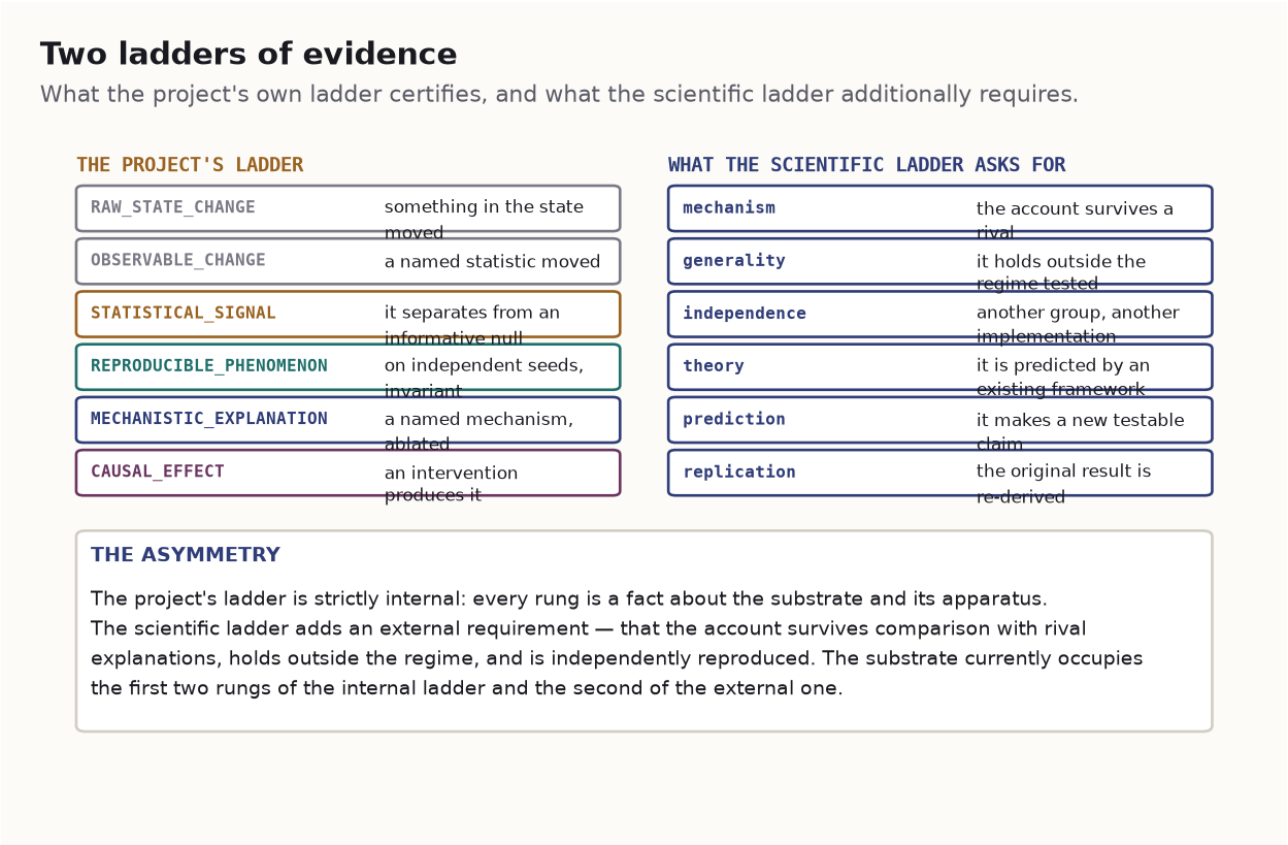


Figure 21. Two ladders of evidence

The project's ladder — raw state change, observable change, statistical signal, reproducible phenomenon, mechanistic explanation, causal effect — is strictly **internal**: every rung is a fact about the substrate and its apparatus.

The scientific ladder adds an external requirement — that the account survives comparison with rival explanations, that it holds outside the regime tested, that another group reproduces it, that it is predicted by an existing framework, that it makes a new testable claim, and that the original result is re-derived.

The substrate currently occupies the first two rungs of the internal ladder and the second of the external one. That is not a criticism of the project; it is an accurate description of a young programme with a well-built instrument and one result. It is also the clearest statement of what would have to happen for the project to count as a scientific contribution rather than a methodological one.

Part VII — What is backed, and what is the project's own proposal

42. A map of the backing

The brief asked to distinguish theories that provide mathematical foundations, describe similar phenomena, share conceptual principles, could complement The Box, differ importantly from it, or could inspire new hypotheses. This chapter does that, and it is deliberately unflattering in places.

42.1 Theories that provide mathematical foundations

Theory	What it provides	Status for The Box
Renormisation group (Fisher; Wilson; Nelson and Fisher)	the mathematics of deriving a global law from local rules, with a fixed point, scaling and universality	`[BACKED]`. The substrate has local rules and a coarse-grained description; it has no fixed point, no ε-expansion and no universality claim. The missing piece is a *block map* — a statement of which aggregate of the fine state is the coarse state
Kolmogorov complexity and algorithmic information (Li and Vitányi; Chaitin)	the bound that turns "generated something not designed" into a measurement	`[BACKED]`. The inequality is unconditional; the substrate's compression test is the application
Topological data analysis (Carlsson; Ghrist)	coordinate-independent signatures of relational structure, and the machinery to compute them	`[BACKED]` for the apparatus. The substrate computes a zeroth persistent homology; it does not claim the barcode means anything
Category theory and toposes (Baez and Dolan; Baez and Hoffnung; Spivak; Fong and Spivak)	the formalisation of relational structure as the primary object, and of "truth" as inhabitation	`[BACKED]` for the programme; `[PROPOSED]` for the transfer to a *dynamic* relational structure
Universal algebra and fixed-point theory (Tarski; Knaster and Tarski)	the existence and construction of stable states in monotone systems	`[BACKED]`. The substrate's fixed points are exactly what the theorem is about; the theorem is about complete lattices and the substrate's states are not presented as one
Skolem's construction and the Löwenheim–Skolem theorem	the behaviour of countable models of uncountable theories	`[BACKED]`. It is a *warning* about global determination claims, and the reason to prefer local derivations
Term rewriting, Church–Rosser, Newman's lemma	the theory of rewriting systems over symbols, with the confluence conditions	`[BACKED]` for the framework; the substrate's rules are contextual rather than syntactic and its system is deliberately non-confluent

Table 5.

42.2 Theories that describe similar phenomena

Phenomenon	Theory	Similarity	Dissimilarity
Structure from local rules	Kac's ring; the renormisation group	a global property derived from a local rule, exactly	Kac's rule is linear and copies; the substrate's rules are nonlinear and create. Kac's geometry is given; the substrate's is not
Synchrony from local coupling	Kuramoto	a critical coupling above which a global order parameter emerges	the substrate has no phases and no natural frequency

Phenomenon	Theory	Similarity	Dissimilarity
Scale-free degree distributions	Barabási and Albert; Jensen	network structure emerging from local growth rules rather than being imposed	the substrate's growth is capacity-limited rather than preferential
Adaptive rewiring	Gross and Blasius	a fitness-dependent probability of targeting a node, producing non-ergodic stationary distributions	the substrate's policy is not fitness-based; it could be made so
Self-sustaining catalytic sets	Kauffman; Hordijk and Steel	a global property (evolvability) computed from a catalytic network as a whole	the substrate's catalysts do not catalyse each other; they are rewritten
Emergent criticality	Bak, Tang and Wiesenfeld; Nagel and Paczuski	a microscopic model with no tuned parameter producing a macroscopic critical state	the substrate's capacity bound is a cap in the configuration, not a conserved quantity
Spatial chaos from deterministic local play	Nowak and May	a deterministic local-rule system producing persistent fluctuating patterns	the substrate has no payoff structure and no players
Emergent locality	Quantum Graphity; Markopoulou and Smolin	a discrete relation structure under local rules, from which locality is derived	Graphity derives *geometry*; the substrate derives identity and lineage
Recurrence and irreversibility from time-symmetric dynamics	Kac's ring; Chandrasekaran and Jain	a finite, time-symmetric microscopic system whose ensemble description is irreversible	the substrate's ensemble is over seeds, not over a time-symmetric measure

Table 6.

42.3 Theories that share conceptual principles

- **Relational quantum mechanics** (Rovelli): a state is relative to another system. The substrate's identity is relative to a relation pattern. The shared principle is that *determination is relational*; the physics has quantum dynamics and probability, the substrate has neither.
- **The categorical and causal-structure literature** (Poulin; Kissinger and Uijlen; Coecke and Lal; Markopoulou): causal structure and time are defined on relational data with no background. The shared principle is that a global order can be *derived* from a local algebra.
- **Stigmergy** (Grassé; Theraulaz and Bonabeau; Heylighen): global order from the trace left by local actions in a shared medium, with no planner. The shared principle is that the medium is the communication channel, and the substrate's medium is its own relation structure.
- **Higher-order network theory** (Battiston and colleagues; Kivelä and colleagues; Benson, Gleich and Leskovec): relations can themselves be related, and the higher-order structure can carry information the pairwise projection does not. The shared principle is that *order ≥ 2 is not decoration* — with the caveat that the substrate's is relational and the literature's mostly simplicial.
- **Neutrally-network evolution** (Wagner; Weiß and Ahnert): a map whose fibres carry hierarchical community structure. The shared principle is that *a partition read off a pattern can be a first-class object with structure of its own*.

42.4 Theories that could complement The Box

- **The renormisation group** would supply the coarse-graining map the substrate lacks, and Israeli and Goldenfeld supply the theorem that the map may exist.
- **Higher-order network statistics and ensembles** (Courtney and Bianconi; Musciotto and colleagues; Chodrow and colleagues) supply the measurement protocol for whether the substrate's order-≥2 structure carries information beyond its dyadic projection.
- **Autocatalytic-set combinatorics** (Hordijk, Steel and Kauffman) supply the vocabulary of enablement and evolvability, and the algorithmic detection that makes the claims checkable.
- **Koopman operator theory** supplies the linearised readout the substrate's spectral layer attempts, with an honest error bar — and the substrate's own measurement shows that bar is large.
- **Graph-rewriting and Petri-net theory** supply the decidability results that bound the substrate's own questions.

42.5 Where The Box differs importantly

Dimension	The Box	The nearest established position	The difference
Dynamics of relations	the relations are rewritten	the toy model, categorical constructions, causal sets, Graphity	the substrate's structure is *dynamic*; the formalisations assume it fixed. This is the single largest difference
Type of relation	relations may have relations as endpoints (order ≥ 2)	simplicial complexes (a clique is one object), hypergraphs, multilayer	relational, not simplicial: a relation *refers* to a relation rather than containing it
Identity	derived from a relation pattern, never stored	stored, or derived in the categorical sense of an equivalence	the substrate's derivation is dynamic and revisable; the categorical one is fixed
Time	several local times, no global clock, no stated consistency	a genuine partial order, with consistency part of the construction	the substrate may not have an order at all. The hypothesis is testable and the test is cheap
Measurement	declared nulls, declared power, one verdict, a diagnosis layer	standard in mature fields; the substrate's version is unusually complete	this is the substrate's *strongest* dimension and the one that needs no defence
Object	no geometry, no metric, no dimension	causal sets and network geometry both derive one	the substrate has no spatial object at all. §37 lists three routes to one
Randomness	none in the dynamics; the ensemble is over seeds	stochastic systems throughout	the second law's applicability to the substrate's ensemble is an empirical question, not a given
Energy	none	every thermodynamic account	the substrate is not a dissipative structure and must not be called one

Table 7.

42.6 Hypotheses in the literature that could inspire new work

- **Israeli and Goldenfeld's coarse-graining result** supplies the licence for the substrate's most ambitious experiment (§33).
- **Kac's ring** supplies the model of what a *derivation* would look like for the substrate — and, in its treatment by Gottwald and Oliver, the warning that the ensemble average and the typical realisation differ, and that the factorisation of the joint distribution is an extra assumption.
- **Baez and Chaudhuri's categorified treatment of feedback in labelled graphs** supplies a formal language in which the *emergent* object of a relational system is the morphism structure rather than the underlying set — which is close to the substrate's own position that the interpretation, not the state, is what is derived.
- **Baez, Pollard, Lorand and Sarazola's coarse-grained conservation laws** supply the existential shape of a finding the substrate could look for (§39).
- **Hordijk, Steel and Kauffman's enablement and evolvability** supply the vocabulary for a global property of a substrate that is not a property of any of its parts.
- **Cook's block encoding of Rule 110** supplies the demonstration that a *self*-coarse-graining of a local rule is possible and yields a new object in the same space with the same dynamics — the closest analogue in the literature to what the substrate's structural campaign is looking for.

43. What the substrate could contribute

43.1 Capabilities that are demonstrated

These are `[IMPLEMENTED]` or `[MEASURED]` in the project's own records and are not claims about the future.

- A relational substrate with derived identity, derived lineage and a shared identifier space, executed deterministically and replayed from a snapshot with an explicit guarantee.
- A possibility space that is finite, exactly enumerable and seed-independent, so that "what could have happened next" is a computable object rather than a sample.
- A measurement discipline: five gates with a fixed precedence, one verdict resolver, a strict separation between *not significant* and *not measurable*, power accounting computed before the run, and a diagnosis layer that attributes a rejection to a named cause.
- A representation-synthesis layer that generates candidate statistics over the substrate in nine families, with a degeneracy audit.
- A structural campaign machinery that varies the rule set and the policy rather than the substrate, with checkpointed, resumable campaigns.
- One structural finding, mechanistically explained, preregistered, confirmed on held-out seeds, with one rival falsified and one left inconclusive.
- One admitted phenomenon, at the mechanistic rung, with ablations that drive the statistic to zero.

43.2 Capabilities that would require further research

Capability	What is missing	What it would take
A derived metric or dimension	the substrate has no spatial object	a spacing functional from relational neighbourhoods, with a triangle-inequality-like constraint and a stable dimension across scales
A coarse-grained description of the birth/fusion balance	no block map exists	a search for an effective rule set of lower descriptive complexity, with complexity measured rather than judged
A universality claim	no exponents, no order parameter	a control-parameter family with a critical point, an order parameter, and a scaling relation
A conservation law for the substrate	none exists	a coarse-grained quantity conserved under the effective rules and not under the primitive ones
A proof of settling or non-settling	impossible internally if the rules are universal	a non-universality argument, or the honest reclassification of the finding as an observation about an undecidable predicate
Derived identity classes with predicted structure	the classes are defined, not modelled	the fibre-partition test and the neutral-network community-structure test
A stigmergy measurement	the architecture is stigmergic but never tested	ablation of trace classes with a published protocol
A causally ordered substrate	the local times are inconsistent or unknown	the transitivity test, and if it fails, an honest statement that the causal-structure programme does not apply

Table 8.

43.3 What would be difficult with other approaches

Four things the substrate's design makes comparatively accessible. Each is a claim about *apparatus*, not about physics.

Testing whether a global order is a coarse-graining. Israeli and Goldenfeld's theorem says such descriptions exist for automata in all classes. Doing that for a given substrate requires enumerating candidate effective rule sets and comparing their descriptive complexity with the primitive ones — a search, not a calculation, and one that needs an instrument that can measure structure at several scales and compare them against matched nulls. The substrate has the instrument; it has not run the search.

Measuring a property that is defined only collectively. Identity classes, lineage depth and the fusion fraction are properties of a relation pattern, not of any object. An ordinary modelling framework would have to *declare* them, and then the question of whether they are informative would be settled by assumption. The substrate derives them, and the derivation can fail.

Keeping the two readings of a null result apart. Every mature field knows the difference between a failed test and an impossible one; few have it *in the pipeline*, with the power computed before the run and the diagnosis attached to the verdict. The substrate does, and the research dossier identifies Gelman and Stern as the citation that makes it a discipline rather than an opinion.

Refusing a claim that a search produced. The campaign that reported UNDERPOWERED rather than a negative result is the practical demonstration that the instrument can be made to say "I could not have detected it". That is rarer than it should be.

44. Eight hypotheses

Each hypothesis is formulated so that it could be wrong, with its scientific foundations, its relation to existing theories, the predictions it would allow, how it could be investigated, what would support it, what would refute it, and its present limitations.

H1 — The derived identity classes are fibres of a well-defined map

Proposes. That the substrate's identity classes are not merely equivalence classes of a relation pattern, but the level sets of a function on a suitable space — and therefore have the internal structure that fibres of maps have.

Phenomenon explained. Why the substrate's identity partition should be *more* than bookkeeping: if the classes are fibres, then the neutral components within each fibre are a second-order structure with its own organisation.

Foundations. The fibre-partition and neutral-network literature (Wagner; Weiß and Ahnert). The higher-order-network finding that community structure can be invisible in a pairwise projection (Benson, Gleich and Leskovec).

Relation to other theories. It would make the substrate's identity commitment a *model* with predictions rather than a definition, and would place the substrate inside the genotype–phenotype-map tradition.

Predictions it would allow. (i) The classes are level sets of a computable function. (ii) The connected components inside each fibre carry hierarchical community structure. (iii) Lineage depth is a computable function of the map.

How to investigate. Construct the candidate map; test (i); apply the established neutral-network community-detection protocol within each fibre; test (iii).

Supporting results. The partition is already computed and already invariant under relabelling, which is a necessary condition.

Refuting results. The classes are not fibres of any well-defined map; or the internal components show no structure above chance against a matched null.

Present limitations. The candidate map has not been written down; the protocol for (ii) assumes a genotype–phenotype map and would need adapting; the substrate's classes are produced by a *process*, not sampled from a population, so the fibre picture is not obviously the right one.

H2 — The relation structure admits a consistent partial order

Proposes. That the substrate's several local times can be made consistent into a partial order on the relation structure — or, failing that, that they demonstrably cannot.

Phenomenon explained. Whether "no global clock" is an *affordance* — a substrate in which time is derived rather than assumed — or merely an *absence*.

Foundations. Causal set theory (Bombelli, Lee, Meyer and Sorkin; Rideout and Sorkin; Dowker and colleagues) and the derived-causal-order programme (Kauffman and Smolin; Markopoulou and Smolin).

Relation to other theories. It is the substrate's route into the literature that treats a partial order on a relational structure as the primary object of physics.

Predictions it would allow. If a partial order exists: a spacing functional becomes meaningful, and the causal-set machinery becomes literally applicable. If it does not: the "no global clock" commitment is an obstruction, and the correspondence is closed.

How to investigate. Build the time-augmented relation graph; test for cycles; if acyclic, test for local finiteness and consistency under the rewriting rules.

Supporting results. The times are attached to relations rather than to the state, which is the necessary form.

Refuting results. Cycles in the time-augmented graph; or the discovery that a relation's time is a rule parameter rather than a property of the pattern.

Present limitations. Cheap and decisive but narrow: it decides one correspondence, not the programme.

H3 — The rule set is universal, and the settling question is therefore undecidable

Proposes. That an explicit universal computation can be embedded in the birth-and-fusion machinery — and that the project's central structural finding is consequently a statement about an undecidable predicate.

Phenomenon explained. The epistemic asymmetry between the two halves of the structural finding, and why no further preregistration will settle it.

Foundations. The undecidability results for cellular automata and physical systems (Wolfram; Perales-Eceiza and colleagues), and Chaitin's completeness result.

Relation to other theories. It would place the substrate inside the class of systems whose long-term behaviour is undecidable.

Predictions it would allow. That no algorithm decides settling for the substrate; that the finding can only ever be empirical; and that the preregistration programme must be reformulated as an empirical claim rather than a theoretical one.

How to investigate. Construct an explicit Gödel numbering of sites and embed a universal machine in the rule and fusion machinery.

Supporting results. The rules are local, discrete and expressive enough to manipulate a payload field; the birth rule creates objects, which is the capability a simulation would need.

Refuting results. No such embedding exists — and then the project has a *theorem* about its settling behaviour, which would be far more valuable than the current empirical finding.

Present limitations. Either outcome is a substantial strengthening; but the experiment is hard to construct and the negative result is hard to certify.

H4 — A coarse-grained description of the birth/fusion balance exists and is much simpler than the primitive rules

Proposes. That there is an effective rule set of strictly lower descriptive complexity that reproduces the large-scale behaviour of the substrate, and that this effective description was not authored.

Phenomenon explained. "The system generated something that was not designed" — in the only form that survives the algorithmic-information bound.

Foundations. Israeli and Goldenfeld's coarse-graining theorem; Cook's block encoding as a self-coarse-graining; the Kolmogorov-complexity bound.

Relation to other theories. The renormisation group, in the substrate's own setting.

Predictions it would allow. (i) An explicit effective rule set. (ii) A measured complexity gap between it and the primitive rules. (iii) The effective description should reproduce the fixed-point/non-settling distinction.

How to investigate. Search over candidate effective rule sets; measure the complexity of each; compare.

Supporting results. The project already measures its statistics at several scales and compares across regimes.

Refuting results. No effective description of lower complexity reproduces the behaviour — in which case the macroscopic description is not a coarse-graining of the microscopic one, and the project is measuring a regime rather than a law.

Present limitations. "Lower descriptive complexity" is not directly computable for arbitrary candidates; a practical surrogate is needed, and the surrogate's own bias would have to be declared.

H5 — The substrate's order-≥2 structure carries information beyond its pairwise projection

Proposes. That the relations-about-relations carry information the pairwise projection loses — and that this is measurable against a matched ensemble with a reported power function.

Phenomenon explained. Whether the substrate's most distinctive representational commitment is substantive or decoration.

Foundations. Benson, Gleich and Leskovec's measurement that higher-order community structure is not recoverable from the pairwise projection; Battiston and colleagues' hypergraph dynamical mean-field framework and its phenomena with no pairwise counterpart.

Relation to other theories. Higher-order network science, to which the substrate would contribute an instance.

Predictions it would allow. A positive mutual-information or z-score against a matched null, with the power reported alongside.

How to investigate. Score the order-≥2 structure against an ensemble matched to the birth/fusion statistics, with the z-score and the power function both reported.

Supporting results. The fusion relation is a relation about relations, and the identity classes are defined by relations among relations.

Refuting results. The order-≥2 structure adds nothing beyond the dyadic projection — in which case commitment 1 should be retired.

Present limitations. The protocol for higher-order structure assumes simplicial or hypergraph structure; the substrate's is relational, and the matching ensemble would have to be constructed.

H6 — The coordination is stigmergic in the strong sense

Proposes. That the substrate's global outcomes are produced by traces stimulating later actions, such that ablating entire *classes* of traces changes the global outcome.

Phenomenon explained. That the substrate's architecture — local rules writing into a shared medium — produces genuine stigmergic coordination, rather than coordination by a different mechanism.

Foundations. The stigmergic lineage (Grassé; Theraulaz and Bonabeau; Heylighen), and the strong form's published definition and protocol.

Relation to other theories. It would place the substrate in the stigmergic-systems literature, which spans ant colonies, distributed computing and institutional design.

Predictions it would allow. (i) Removing a class of traces changes the outcome distribution. (ii) The effect is not reproduced by removing individual relations of the same count.

How to investigate. Ablate classes of traces and measure the change in the outcome distribution; compare against the matched single-relation ablation.

Supporting results. The rules read relations and write relations into a shared medium; there is no planner and no central control.

Refuting results. Class ablation and relation ablation have the same effect, which would show the coordination is not trace-mediated.

Present limitations. The reserved high tag range means that *some* of the medium's content comes from outside the system, which is a genuine qualification of the strong form and should be tested explicitly.

H7 — A conservation law exists at the coarse-grained level only

Proposes. That some quantity is conserved under an effective description of the rule set but not under the primitive description.

Phenomenon explained. A structural analogue of the substrate's own derived-identity result: a global property that holds only once the description is coarse enough.

Foundations. The categorical result that conservation laws can hold at the coarse-grained level and fail at the level of the components (Baez, Pollard, Lorand and Sarazola); Noether's theorem as the contrast case, with its hypotheses made explicit by Palais and by Anderson and Pohjanpelto.

Relation to other theories. Category theory and the physics of conservation laws; and the substrate's own invariance battery as the search instrument.

Predictions it would allow. A conserved quantity of the effective rules that is not conserved by the primitive rules, and a demonstrable dependence on the coarse-graining.

How to investigate. Compute candidate quantities along trajectories; search for one that is constant under an effective rule set and not under the primitive rules.

Supporting results. The substrate already computes coarse-grained descriptions of its own behaviour at several scales and compares them.

Refuting results. Every candidate quantity that is constant at a coarse scale is also constant at the primitive scale, or none is constant at any scale.

Present limitations. Noether's theorem does not apply to the substrate at all — there is no action functional and no differentiable group — so the search is a heuristic with a published precedent,

not a theorem.

H8 — The identity of the derived classes has a measurable information content, and the selection of one site per step is an information-consuming act

Proposes. That the entropy of the identity partition, and the information content of the policy's selection among eligible sites, are well-defined and measurable quantities for the substrate — and that the selection is the substrate's symmetry-breaking-like event.

Phenomenon explained. What the substrate's "one site per step" costs in information terms, and what the derived partition is worth.

Foundations. Shannon entropy over the partition; the general shape of "symmetry breaking as a reduction in the number of distinguishable states"; and the declared gap that no primary source states the sharp version of the latter, which is itself a reason to construct it here.

Relation to other theories. Statistical mechanics' thermodynamics of phase transitions; the substrate's own ensemble discipline.

Predictions it would allow. A measurable entropy of the identity partition as a function of the step; a measurable information cost per selection; and a comparison between the two.

How to investigate. Compute the partition entropy over the derived classes; compute the entropy of the policy's distribution over eligible sites; measure both against the regime transition.

Supporting results. The substrate already computes entropy over depth histograms, component-size profiles and identity-class sizes, so the estimators exist.

Refuting results. The quantities are not stable across regimes, or carry no information about the transition, which would make the symmetry-breaking analogy empty.

Present limitations. "Information consumed by a selection" is not a thermodynamic quantity without a physical realisation, and the document's own negative findings (§32) warn against dressing the substrate's bookkeeping as physics.

45. The great open questions

1. **Is the substrate's relation structure a partial order?** (§16.4, H2) — the single highest-value experiment in this document.
2. **Is the rule set universal?** (§38, H3) — decides whether the project's one structural result is an empirical discovery or a restatement of the halting problem.
3. **Does a coarse-grained description of the birth/fusion balance exist?** (§33, H4) — the only rigorous form of "generated something not designed".
4. **Does the order-≥2 structure carry information beyond the dyadic projection?** (H5) — decides whether the substrate's most distinctive commitment is substance or decoration.
5. **Is the coordination stigmergic in the strong sense?** (H6) — the cleanest published protocol for the substrate's most distinctive architectural claim.
6. **Is the substrate's relation structure a poset *and* does it admit a spacing?** (§37) — the route to a derived metric, without which the whole emergent-geometry correspondence stays metaphorical.

7. **Can the project's own power accounting be made rigorous?** — specifically, whether its null model is exchangeable under the very birth and fusion rules being tested. If it is not, every finding is a measurement artefact, and this is the one question that threatens the whole apparatus.
8. **What is the substrate's relation to the reserved high tag range?** — formally, whether the high range is inert or a distinguished sub-vocabulary with a different provenance, and whether treating a high-range tag as an externally supplied trace changes the coordination.

Part VIII — Closure

46. Conclusion

The question this investigation was asked is whether there is a deep connection between the principles of The Box and some of the largest open questions in science, and the answer has three layers, and the layers are the finding.

The first layer: the substrate's commitments are already the commitments of established science. A relational structure in which identity is derived rather than stored is a structural idea with a long history in category theory, algebraic topology and network science — and the toy model is the strongest formal statement of the intuition that geometry can be relational. Local rules whose global statistics are tested against nulls is the central methodological move of statistical physics. A statistic required to survive a change of measurement context is the operational question that Kochen–Specker made decisive. Several local times with no global clock is the posture the problem of time forces on quantum gravity. None of this is exotic, and none of it is the project's invention.

The second layer: one or two connections carry real formal weight, and they are testable. That coarse-grained descriptions exist for cellular automata in *all* behavioural classes, including universal and therefore undecidable ones, and that the probability of finding one approaches unity as the scale coarsens — that is a theorem-backed version of "the system generated something that was not designed", and it comes with a ceiling in the same paper. That the identity classes of a substrate might be the fibres of a map, with the neutral components inside each fibre carrying hierarchical community structure, is a formal object the literature already knows how to measure. That a metric can be *derived* from a locally finite partial order is a theorem rather than a conjecture, and the substrate's relation structure might be one — a question that a cycle test on the time-augmented graph would answer for the price of an afternoon.

The third layer, and the one that matters most: what the project has actually built is a discipline, and the discipline is the contribution. The gates, the single verdict resolver, the strict separation between *not significant* and *not measurable*, the power computed before the run, the diagnosis that attributes a rejection to a named cause, the preregistration with held-out seeds, the falsified rival and the inconclusive rival left standing, the four replay guarantees, the divergence attribution, the persisted analysis flags. Every one of these has a precedent in a mature discipline, and Gelman and Stern's paper is the reason the *not measurable* verdict is a discipline rather than a private convention. But assembling them into one pipeline, and then having the pipeline refuse its own underpowered campaign rather than report it as a negative result, is not something the literature hands out.

And the honest bottom line. One phenomenon is admitted, at the mechanistic rung, with ablations that drive the statistic to zero. One structural finding is confirmed on held-out seeds, with a rival falsified and a rival left unresolved. Both are about *one rule set* on *one substrate*. The project's own verdict — mechanism explained, no discovery — is the correct summary, and this

document does not improve on it.

What would improve on it is stated as eight hypotheses and eight open questions in §44 and §45, and the first two of those questions — *is the relation structure a partial order* and *is the rule set universal* — are cheap, decisive, and would each change the epistemic status of the project's one result. The programme's next two weeks are worth more than its next two years of feature work, and the reason is not aesthetic: two of those questions can make the existing result into a theorem or into a restatement of the halting problem, and the project cannot currently tell which.

47. The limits of the instrument

This chapter is about the limits of *computational research as a method*, not a list of historical bugs. The point is that a computational instrument has a characteristic shape of blindness, and that shape has to be named before its results are read.

47.1 A model is not the world

Every result in this document is a result about a discrete, finite, deterministic relational system. It is not a result about a physical, biological or cognitive system, and no amount of computational sophistication changes that. The step from the model to the domain is a *modelling* step, and it carries its own assumptions, its own idealisations and its own failure modes, none of which the instrument can audit for itself.

47.2 The representation decides the question

The substrate can only measure what its representation makes measurable. It has fifty-three hand-written observables and synthesises thirty more, and the synthesis experiment showed that the substrate supports a far larger measurement space than the hand-written bank uses — so the *hand-written* set is a limitation, not a ceiling. But the synthesis was itself a search over families that a human chose. The question of whether the chosen families are the right ones is not answerable from inside the instrument.

47.3 Abstraction loses information, and the loss is not recoverable

Going from a rich structure to a statistic is a projection. The statistics that survive projection are the ones that were projected onto; the ones that are not visible are not thereby absent. This is the deep reason the null-model discipline matters: a null that does not preserve the relevant properties turns structure into artefact, and no amount of statistical care afterwards repairs it.

47.4 Complexity is not understanding

A trajectory can be long, complicated and incompressible without being understood. The Kolmogorov bound of §40.1 is the sharpest statement of the limit: what the substrate produces is describable, and the interesting question is whether its description is *short*. A short description is not the same as an explanation — it is a necessary condition for one, and the instrument supplies the first, never the second.

47.5 Emergent behaviour is hard to interpret, and harder to generalise

When a global property appears, two questions must be answered before it means anything: *is it a consequence of these rules or of this configuration*, and *does it survive a change of regime*. The substrate's apparatus answers the first with null models and the second with the invariance battery, and both answers can be "no". A behaviour that appears in one regime and evaporates in another is not a law, and the instrumentation's job is to catch that — which it does, and which is why the project's campaigns so often report a null.

47.6 Prediction is limited by computability, not by effort

§40's ceiling is not a methodological inconvenience that more compute would remove. For a rule set capable of universal computation, the settling question is undecidable in principle. Any instrument of this kind inherits that limit, and the honest response is to declare it rather than to keep searching for a better search.

47.7 The instrument cannot certify itself

The gates test whether a result survives *within* the apparatus. Nothing in the apparatus can verify that the apparatus measures what it claims to measure. The closest available check is external: a second implementation, a second group, a different protocol. The project has a second implementation at fidelity level 3 — structurally equivalent, not byte-compatible — and that is the beginning of an external check, not the end of one.

47.8 The apparatus can be wrong about its own apparatus

The project's own history is the evidence: a statistic that was algebraically identical to the node count, a verdict vocabulary with a documented but unreachable verdict, a predicate measuring age rather than starvation, a serialisation that silently dropped a bound, a test suite whose counts were never reconciled between documents. None of these was found by running the code; all were found by *auditing* the code, and the instrument that found them was a different one — a human reading, or a separate analysis pass. This is the deepest practical limit of the whole enterprise, and it is the one the project has been most honest about.

48. Glossary of scientific terms

Aggregate / macrostate. A property of a collection of elements that is defined over the collection as a whole and typically compresses information about the individual elements.

Autocatalytic set. A set of catalysts each of which is catalysed by another element of the set; irreducible if no proper subset is so.

Automorphism / relabelling equivariance. A property of a system whose observable behaviour is unchanged under a permutation of the identifiers.

Bell number. The number of partitions of a set of n elements; the number of ways the elements can be grouped into identity classes.

Betti number. The k-th Betti number of a space counts the k-dimensional holes; for a graph, β_0 is the number of connected components and β_1 the cycle rank.

Category / dagger / topos. A category with objects and morphisms; *dagger* means every morphism has a converse; a topos is a category with a logic of its own, in which truth is

inhabitation rather than a value.

Church–Rosser property (local confluence). A rewriting system is confluent if every two one-step divergences can be joined; local confluence plus termination gives confluence (Newman's lemma).

Coarse-graining. A map from a fine description to a coarser one that preserves the behaviour of interest; the renormisation group is a systematic theory of such maps.

Configuration space. The space of all configurations of a system; the setting in which an action functional and a variational structure are defined.

Contextuality. The dependence of the value assigned to an observable on which set of compatible observables is measured alongside it.

Causal set. A locally finite partially ordered set with a measure, from which a continuum Lorentzian geometry is derived.

Derivation / weak emergence. A macrostate is weakly emergent if it can be derived from the microdynamic and the external conditions *but only by simulation* (Bedau); strong emergence if it is not deducible even in principle (Chalmers).

Dichotomy. A distinction whose two branches are exhaustive and mutually exclusive.

Dissipative structure. A non-equilibrium ordered state maintained by a flow of energy from the surroundings, sustained by nonlinear amplification of fluctuations (Prigogine).

Emergence, strong and weak. See *derivation*. The distinction is between two readings of the word and no physical phenomenon has been demonstrated to be strongly emergent.

Enablement. An autocatalytic catalyst that is produced by the set but not currently used by it; it measures the set's latent capacity to grow.

Entanglement, monogamy. Entanglement cannot be freely shared: if two systems are maximally entangled, a third shares none of it.

Evolvability. How many new catalysts can be added to an autocatalytic set while keeping it self-sustaining; the size of its adjacent possible.

Fibre. The preimage of a point under a map; a generalisation of a connected component.

Fixed point. A state that maps to itself. In a monotone system on a complete lattice, fixed points are guaranteed (Tarski).

Fibre partition / neutral network. The connected components of a level set of a continuous map; sets of genotypes with identical phenotype.

Holomorphism. A map between algebraic structures that preserves the operations.

Homomorphism (relational). A map between relational structures preserving the relations; the formal content of the substrate's relabelling invariance.

Invariance (measurement). The property of a statistic's value being unchanged under a specified change of measurement context — relabelling, window shift, policy change.

Kolmogorov complexity. The length of the shortest program producing a string; $K(f(x)) \leq K(x) + K(f) + O(1)$.

Louvain method. A heuristic for community detection by maximising modularity; deterministic when tie-breaking is fixed.

Modularity. The fraction of edges within communities minus the expectation in a degree-preserving randomisation; a measure of community structure in a network.

Noether's theorem. A differentiable symmetry of an action functional yields a conserved current along solutions. It is *not* the claim that an invariant statistic implies a conserved quantity.

Order parameter. A quantity distinguishing the phases of a system, typically vanishing in one phase and non-zero in the other.

Partial order. A relation that is reflexive, antisymmetric and transitive; the causal-set programme's primary object.

Phase transition. A change of qualitative behaviour as a control parameter crosses a critical value; in the substrate's language, a regime change.

Power. The probability that a test detects an effect of a given size; the substrate computes the resolution floor and declares results untestable when the budget cannot express the threshold.

Preregistration. The fixing of a hypothesis, its protocol and its analysis before the data are seen.

Quotient space. The space of equivalence classes of a relation; the substrate's identity classes live in one.

Rennormalisation group. The flow of effective couplings under coarse-graining, with fixed points characterising universality classes.

Resolution floor. The smallest p-value a null budget can express, $1/(n+1)$.

Riemannian / spectral dimension. A characterisation of how many independent directions a structure exhibits; a spectral dimension can be read off a relational structure.

Robust chaos. Chaos that persists under parameter perturbation, as opposed to chaos that is a knife-edge accident.

Self-organised criticality. A system with no external tuning that drives itself to a critical point; its status in the sandpile family is contested.

Spectral concentration test. A test for whether a Koopman spectrum is concentrated in a few modes, used as a proxy for closure.

Stigmergy. Coordination through a shared medium, in which the trace left by each action stimulates subsequent actions, without a planner.

Strong emergence. See *derivation*.

Topological signature. A coordinate-independent summary of the shape of a structure, such as a persistent-homology barcode.

Undecidability. The property of a class of questions for which no algorithm exists that decides all instances.

Universal computation. The property of a system able to simulate an arbitrary Turing machine; its consequence is that long-term behaviour is undecidable.

Verdict, in the substrate's vocabulary. `EXPLAINED`, `SUPPORTED`, `UNTRANSLATED`, `REJECTED`, `UNTESTABLE`, `CAUSAL`; the last is a statement about the apparatus, not about the world.

Weak emergence. See *derivation*.

49. Sources and references

Every reference below was checked against a publisher's page, an archive record, an authoritative reference work, or the reference list of a paper that was itself opened. Where an identifier could not be independently confirmed it is marked as unverified rather than reconstructed. Preprints are labelled as preprints. Several plausible-looking identifiers were found to resolve to entirely different papers and have been excluded; those are listed at the end, because a document that only reports its successes misrepresents its own diligence.

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49.5 Identifiers that were checked and excluded

These circulate in citation lists and were found to resolve to entirely different works. They are recorded because a bibliography that silently drops them would give a false impression of completeness.

Identifier	Actually resolves to
10.1038/249328a0	Barbour, *Relative-distance Machian theories* — **this is the paper cited**; it is not a paper titled "Absolute time"

Identifier	Actually resolves to
10.1115/1.1553433	a *book review* of Wolfram's *A New Kind of Science* in Appl. Mech. Rev. 56, B18 (2003) — not the book
10.1098/rstb.2008.0096	a speciation paper — not the emergence introduction the identifier is usually attached to
10.1103/PhysRevLett.71.467	Koshibae et al. — not a self-organised-criticality paper
10.2307/2268983	a 1949 *J. Symbolic Logic* review — not the Kramers–Wannier duality paper
10.1103/RevModPhys.46.597	Fisher, *Scaling and Renormalization in Statistical Physics* — **not** Wilson
10.1103/RevModPhys.77.137	Acebrón et al., the Kuramoto review — **not** Strogatz
10.1016/j.physrep.2015.10.008	Rodrigues, Peron, Ji & Kurths — **not** Wiley, Brunel & Strogatz
10.1038/269759a0	Zahler & Sussmann — **not** Zeeman
10.1103/PhysRevE.51.2909	Nagel & Paczuski — **not** Helbing

Table 9.

Four arXiv identifiers that were recalled from memory were found to belong to unrelated papers and are not used: 1007.4463, 1711.01261, 1603.06548, 1903.09638.

49.6 Works named in the brief that could not be located

Listed so that they are not mistaken for an absence of literature. In each case a verified substitute was used instead.

The Baez–Dolan toy-model paper was located by the mathematics dossier but its identifiers could not be independently confirmed; the toy model is therefore described in the text and its two canonical secondary anchors are cited instead. The Fischer–Renner paper of 1972, the Nowak–Bonhoeffer–May "game of life" contribution, Konopkowski's causal-jár paper, the Hojhati–Short–Smolin caveats, the Block–Gefen–Kohlmeyer–Zhong RGCA hierarchy, Anderson's 1969 paper, Villari's 2008 paper, Gotay–Isenberg–Marsden, Kac's 1956 original, de Bruijn's morphic sequences, Sipser on Kolmogorov complexity, Schmidhuber's Gödel machines, Koopman's 1931 primary source, Wootters' concurrence paper, the two AHMS papers, Engelhardt–Wallace, Chong–Shenker–Stanford, 't Hooft's original holography proposal, and the Seiberg–Tingle ER=EPR paper were not located with a verifiable identifier. The Ravetto objection to relational quantum mechanics, and the formulation of the "no-signalling problem" in that framework, are stated in the project's own literature only by the name of the objection; the verified adjacent material is cited instead and the gap is recorded rather than papered over.

49.7 A note on the coverage of this bibliography

The reference lists behind this document were assembled by targeted retrieval against the arXiv, Crossref, OpenAlex and INSPIRE records and the Stanford Encyclopedia of Philosophy, with every identifier re-resolved before use. That method is **a lower bound on the relevant literature rather than a census**. Rate limiting on the arXiv API, and the fact that the discovery paths were reliable only for authors that could be named in advance, mean that papers by unconsidered authors, or in unscanned categories, could exist and be missed. The corpus is therefore best read as *sufficient for the claims made*, and not as exhaustive.